



Continuity of the Positioning Service

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Outlines



Context

Outdoor Positioning: existing and future solutions

Indoor Positioning: existing and future solutions

Continuity of the Positioning Service

Conclusion

Problematics - Contexts



- For Outdoor Location
 - Satellite Positioning System GNSS, GPS, Glonass, Galileo
 - Works with a public accuracy between 5 and 20 meters (1 frequency) in open horizon condition
 - Difficulties to work in a partial horizon environment
 - Does not work in a closed environment
 - Difficulties to start according to condition (Hot, Warm, Cold start)
- For Indoor Location
 - Good Propagation model for homogeneous area
 - Buildings are heterogeneous area
 - Accuracy inside an heterogeneous area is a hard task to estimate
 - Accuracy can change according to some modifications

Outdoor location: GPS Architecture

- Ground segment
 - 4 Ground units which allow to communicate with all the satellites in space and a master control.
 - Send Almanac (long term) and Ephemeris (short term) data, 50bps
 - Measure accurately satellite trajectories
- Spatial segment
 - 24 satellites in orbit on 6 planes
 - Fifth generation of satellites
- User segment
 - Mass market receivers (L1 frequency), mainly smartphone
 - Professional receivers (L1 and L2 frequencies)
 - Receivers with correction SBAS or D-GPS
 - Coordinates calculus using the GNSS hardware interface

GPS: Ground segment

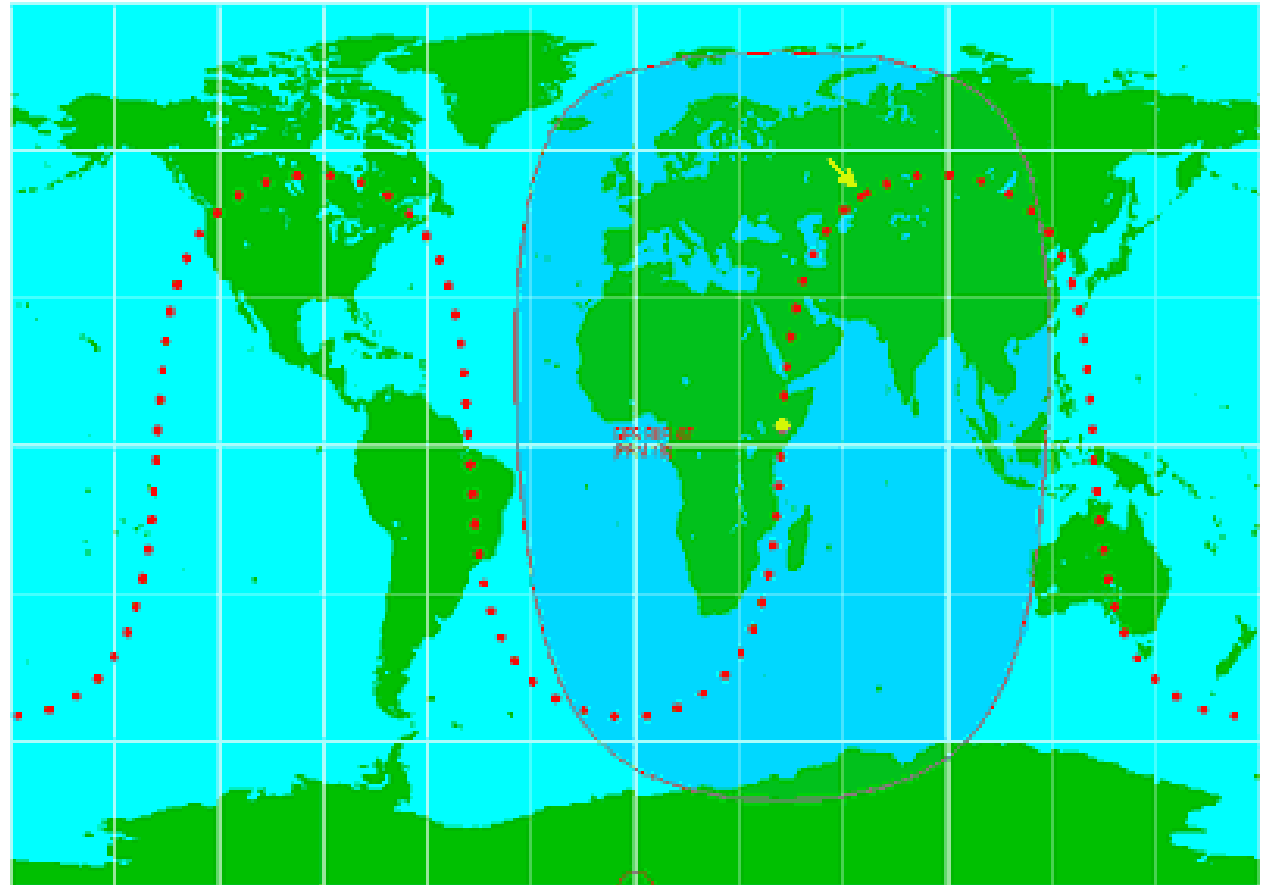
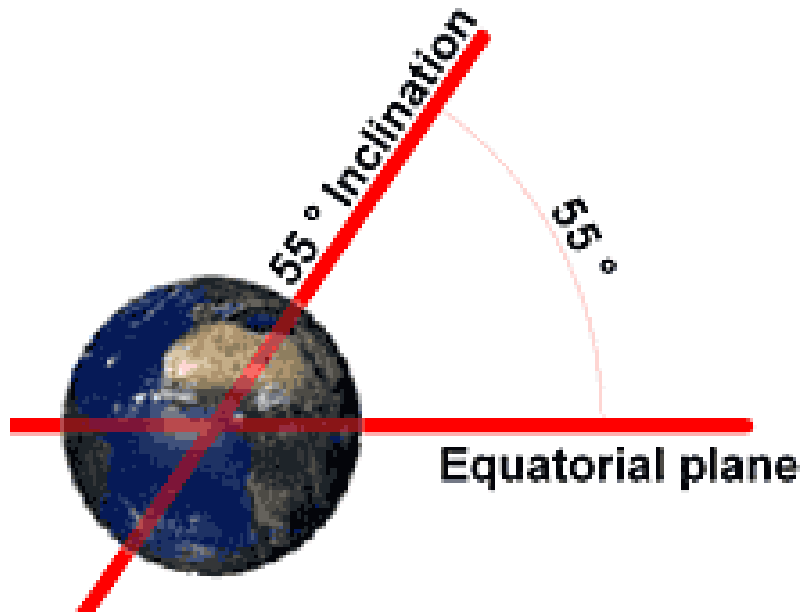


Global Positioning System (GPS) Master Control and Monitor Station Network

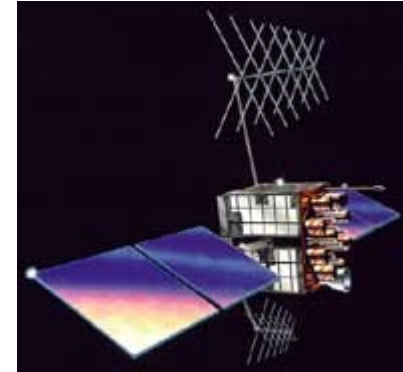
GPS: Spatial topology



- 6 orbits with 55° of inclination



Satellites Specification



Generation Block I (1978-85)
designed for 5 years - 845kg
Solar panel 400W
Nickel-cadmium batteries
Hydrazine engine
Atomic clock: 1 with cesium
and 2 with rubidium

Generation Block IIR (2005-)
designed for 15 years - 2 tons
Solar panel 750W
Nickel-cadmium batteries
Hydrazine engine
Atomic clock (10-13s): 3 with
rubidium
Signal power L1-L2 : 50W
Appearance of the C/A code
(Coarse acquisition)

User Segment



Chipset	Sirf Star 3	MTK	GPSone
Sensitivity	-159dBm	-165dBm	-160dBm
Time to First Fix	Cold 60s Warm 38s Hot <8s (max 2h)	Cold 36s Warm 33s Hot 1s (max 2h)	Cold 35s Warm 29s Hot <8s (max 2h)
NMEA	GLL GLA RMC VTG GSV GSA	GLL GLA RMC VTG GSV GSA ZDA	GLL GLA RMC VTG GSV GSA
Calculus frequency	1Hz	1 to 5Hz	1Hz
L1 channel	20	32	32

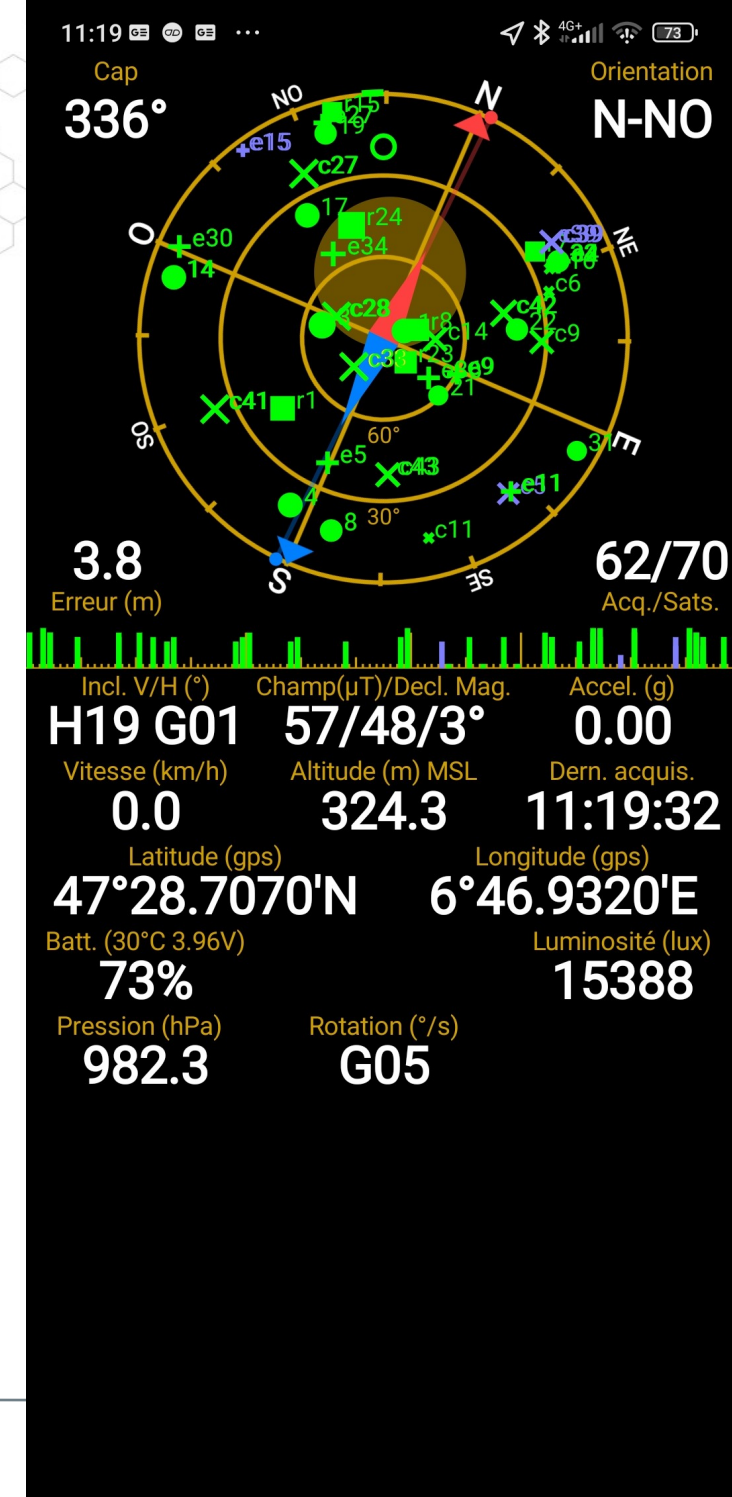
User Segment - Smartphone

- Samsung Galaxy S8,
- Chipset Qualcomm Snapdragon 835
- Application: GPS Status



User Segment - Smartphone

- First smartphone with 2 frequencies L1 and L5
- Chipset Qualcomm Snapdragon 855
- Application: GPS Status



L1 and L2 signals



L1=1575.42MHz (154 x 10.23 MHz)

BPSK signal at 1.023 MHz

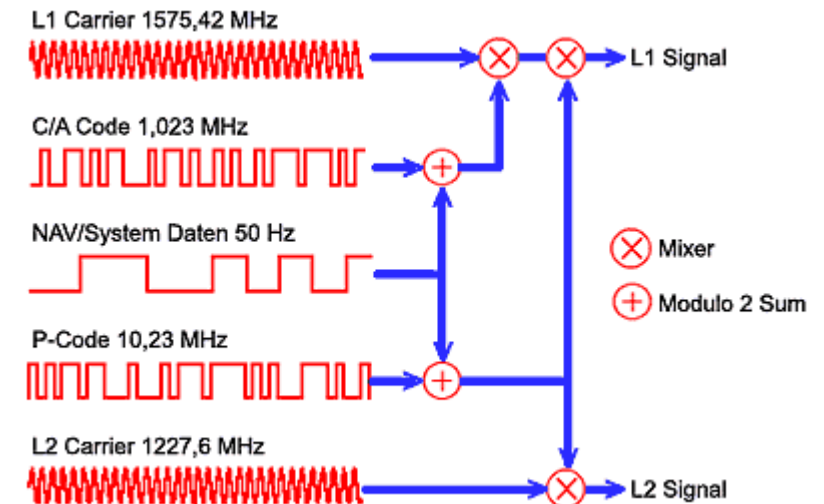
C/A code (Coarse Acquisition), 1023 bits

Pseudo Range Noise (PRN), satellite id

L2=1227.60MHz (120 x 10.23 MHz)

BPSK signal at 1.023 MHz

P-code is more accurate than C/A code and is crypted combined with W-code producing the Y-code (P(Y)).





The image shows the Intel ZED-F9P camera module. It is a white rectangular board with a black border. The Intel logo is visible in the top left corner, and the text "ZED-F9P" is printed in the center. The board is populated with various electronic components, including a large black heat sink on the right side.

One of the first
L5 chipset receivers

Neumann-Hoffman code to reduce the synchronisation time by reducing the search complexity of the synch. algorithm (phase offset and Doppler Effect)

[illegible]

Coordinates calculus method



- By receiving, Almanac (long term) and Ephemeris (short term) data, the user device knows the exact satellites coordinates over time
- Measuring reception time of satellite signals, the pseudo-range distances are calculated
- By combining at least, 4 satellites coordinates and 4 pseudo-range distances, the 3D-coordinates of the device can be calculated
 - 3 unknowns are 3D-coordinates
 - 1 unknown is the synchronization error between user clock device and synchronized clocks of satellites

Overall Error of accuracy



- ~5m due to the ionosphere
 - ~2.5m due to the satellites coordinates
 - ~2m due to the time synchronization between satellites
 - ~1m due to multipath
 - ~0.5m due to troposphere
 - ~1m due to calculus
-
- Overall error of 10 to 15m in the very best condition of reception using L1 frequency only.



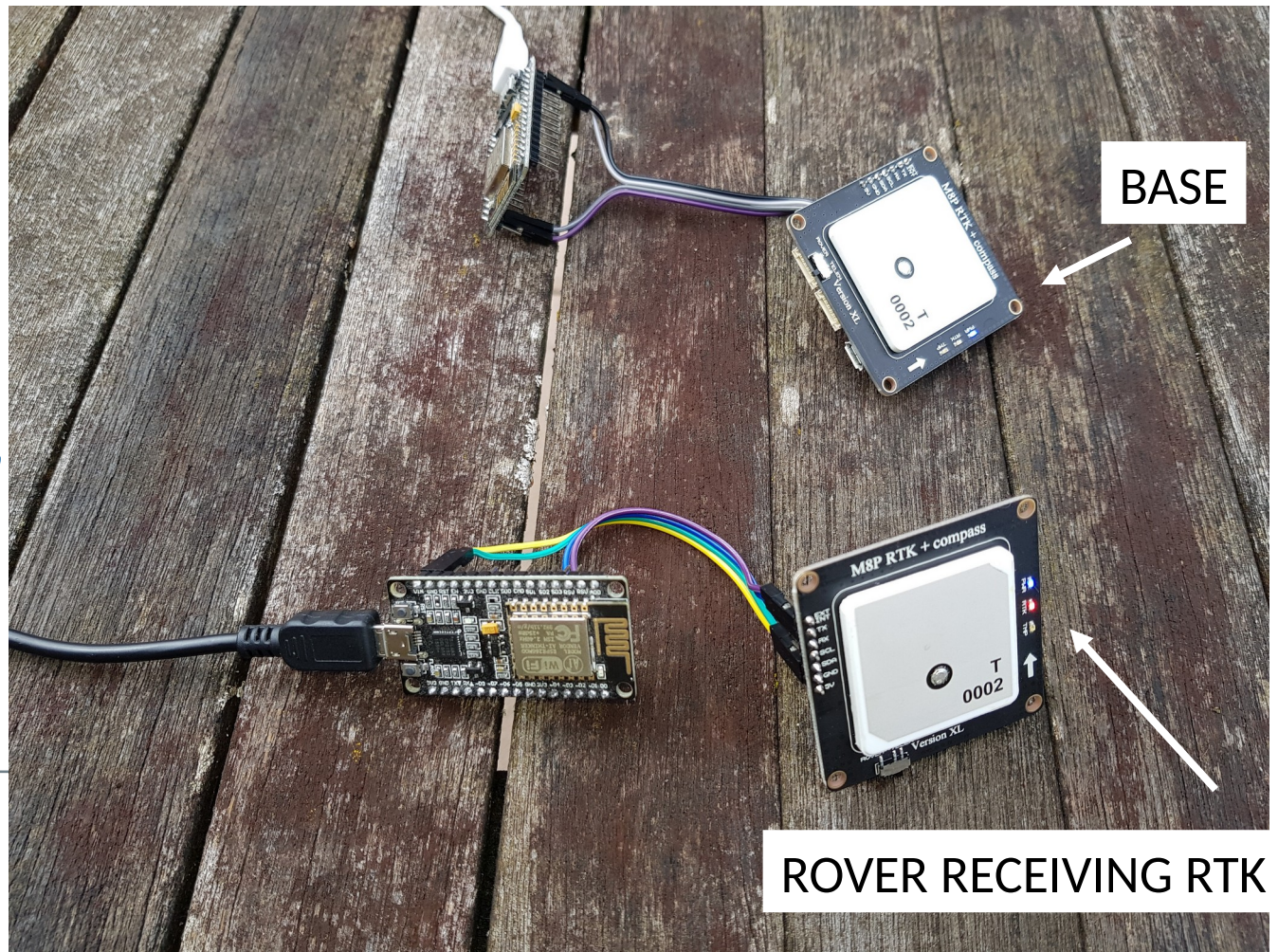
- Very fast Start - Assisted-GNSS
 - Data are sent by Internet on smartphone (excepted when Data Roaming is off)
- Integrity function
 - RAIM: receiver autonomous integrity monitoring
 - Specific function included in aeronautic and marine devices
 - Redundant calculus using different set of satellites in order to exclude potential erroneous transmission from one satellite or fake satellite trying to send malicious errors
- Feedback systems
 - By using ground unit receiving satellite signals and comparing calculated coordinates with true coordinates
 - After measuring corrections, these must be send to user device in order to improve accuracy and precision



- Feedback systems
 - **SBAS: Satellite-based Augmentation System**
 - Free regional improvement: WAAS (USA), EGNOS (Europe), GAGAN (India), MSAS (Japan)
 - Ground stations at known coordinates receive satellites signals and calculate a correction time for each satellite in order to compensate the reception condition as fast as possible
 - Corrections are sent by geostationary satellites and Internet
 - **Differential GPS and Differential GNSS**
 - Private ground station located near from the user device in order to measure the very local atmospheric conditions
 - Direct communication link between the ground station and the user device in order to have real time correction
 - A posteriori calculus in order to stabilize the results

Low-cost Differential GPS (D-GNSS)

- Example of D-GNSS to fix the coordinates of outside ground station with a relative accuracy of 5cm.
- D-GNSS Chipset
u-blox NEO M8P
- *BASE* sends RTK correction to Rover via Wi-Fi/IP/TCP connection





```

Packet Console
11:55:54 R -> RTCM3 1077, Size 174, 'GPS MSM7'
11:55:54 R -> RTCM3 1087, Size 130, 'GLONASS MSM7'
11:55:54 R -> RTCM3 1005, Size 25, 'Stationary RTK reference station ARP'
11:55:55 R -> NMEA GNRMC, Size 68, 'Recommended Minimum Specific GNSS Data'
11:55:55 R -> NMEA GNVTG, Size 35, 'Course over Ground and Ground Speed'
11:55:55 R -> NMEA GNGGA, Size 76, 'Global Positioning System Fix Data'
11:55:55 R -> NMEA GNGSA, Size 63, 'GNSS DOP and Active Satellites'
11:55:55 R -> NMEA GNGSA, Size 57, 'GNSS DOP and Active Satellites'
11:55:55 R -> NMEA GPGSV, Size 70, 'GNSS Satellites in view'
11:55:55 R -> RTCM3 1077, Size 174, 'GPS MSM7'
11:55:55 R -> UBX MON, Size 8, 'Monitor'
11:55:55 R -> UNKNOWN ???, Size 141, 'UNKNOWN'
11:55:55 R -> NMEA GNGSA, Size 57, 'GNSS DOP and Active Satellites'
11:55:55 R -> NMEA GPGSV, Size 69, 'GNSS Satellites in view'
11:55:55 R -> UNKNOWN ???, Size 63, 'UNKNOWN'
11:55:55 R -> NMEA GPGSV, Size 55, 'GNSS Satellites in view'
11:55:55 R -> NMEA GLGSV, Size 70, 'GNSS Satellites in view'
11:55:55 R -> UBX MON-VER, Size 258, 'Version'
11:55:55 R -> RTCM3 1077, Size 174, 'GPS MSM7'
11:55:55 R -> RTCM3 1087, Size 130, 'GLONASS MSM7'
11:55:55 R -> RTCM3 1005, Size 25, 'Stationary RTK reference station ARP'
11:55:57 R -> NMEA GNRMC, Size 68, 'Recommended Minimum Specific GNSS Data'
11:55:57 R -> NMEA GNVTG, Size 35, 'Course over Ground and Ground Speed'
11:55:57 R -> NMEA GNGGA, Size 76, 'Global Positioning System Fix Data'
11:55:57 R -> NMEA GNGSA, Size 63, 'GNSS DOP and Active Satellites'
11:55:57 R -> NMEA GNGSA, Size 57, 'GNSS DOP and Active Satellites'
11:55:57 R -> NMEA GPGSV, Size 70, 'GNSS Satellites in view'
11:55:57 R -> UNKNOWN ???, Size 254, 'UNKNOWN'
11:55:57 R -> NMEA GNGSA, Size 63, 'GNSS DOP and Active Satellites'
11:55:57 R -> NMEA GNGSA, Size 57, 'GNSS DOP and Active Satellites'
11:55:57 R -> UNKNOWN ???, Size 187, 'UNKNOWN'
11:55:57 R -> NMEA GLGSV, Size 24, 'GNSS Satellites in view'

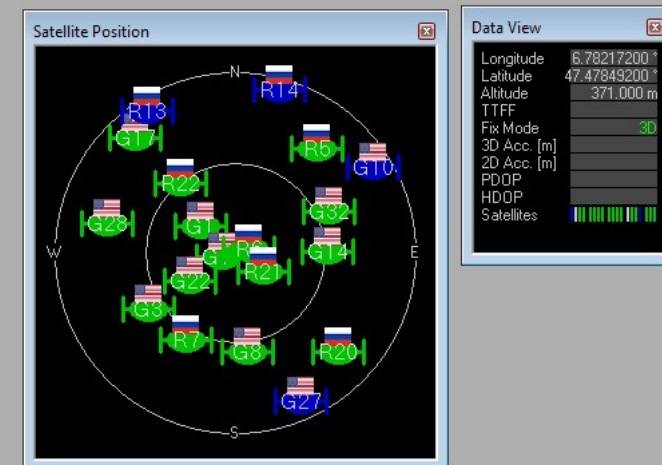
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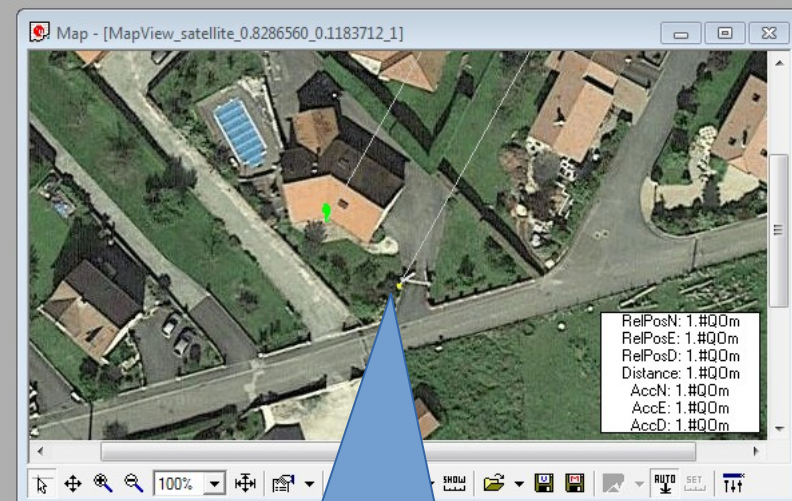
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11:55:55 $GNGSA,A,3,85,70,86,71,84,69,,,,,99.99,99.99,99.99*2D
11:55:55 $GPGSV,3,1,11,01,68,307,49,03,38,237,46,08,40,173,44,10,09,058,34*7A
11:55:55 $GNGSA,A,3,85,70,86,71,84,69,,,,,99.99,99.99,99.99*2D
11:55:55 $GPGSV,3,1,11,01,68,307,49,03,38,237,46,08,40,173,44,10,09,058,33*7D
11:55:55 $GPGSV,3,3,11,27,09,156,28,24,283,41,32,39,066,47*4F
11:55:55 $GLGSV,3,1,09,69,24,038,39,70,83,073,39,71,40,210,42,77,07,328,13*6F
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11:55:57 $GNVTG,,T,,M,0.000,N,0.000,K,A*3D
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11:55:57 $GNGSA,A,3,85,70,86,71,84,69,,,,,99.99,99.99,99.99*2D
11:55:57 $GPGSV,3,1,11,01,68,307,49,03,38,237,46,08,40,173,44,10,09,058,33*7D
11:55:57 $GNGSA,A,3,11,01,22,08,14,32,03,17,28,,99.99,99.99,99.99*2E
11:55:57 $GNGSA,A,3,85,70,86,71,84,69,,,,,99.99,99.99,99.99*2D
11:55:57 $GLGSV,3,3,09,,,41*69

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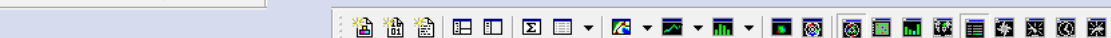
U-Center Software



Data View	
Longitude	6.78217200
Latitude	47.47849200
Altitude	371.000 m
TTFF	
Fix Mode	3D
3D Acc. [m]	
2D Acc. [m]	
PDOP	
HDOP	
Satellites	



RelPosN: 1.#Q0m
 RelPosE: 1.#Q0m
 RelPosD: 1.#Q0m
 Distance: 1.#Q0m
 AccN: 1.#Q0m
 AccE: 1.#Q0m
 AccD: 1.#Q0m



Messages - CUSTOM

NMEA
RTCM3
UBX
UNKNOWN
CUSTOM

CUSTOM

812 s

Configure - Time Mode 3

LOGFILTER (Log Settings)

MSG (Messages)

NAV5 (Navigation 5)

NAV5S (Navigation Expert 5)

NAV5S (Navigation Expert 5)

NAV5S (Navigation Expert 5)

NAV5S (Navigation Expert 5)

NAV5S (Navigation Expert 5)

NAV5S (Navigation Expert 5)

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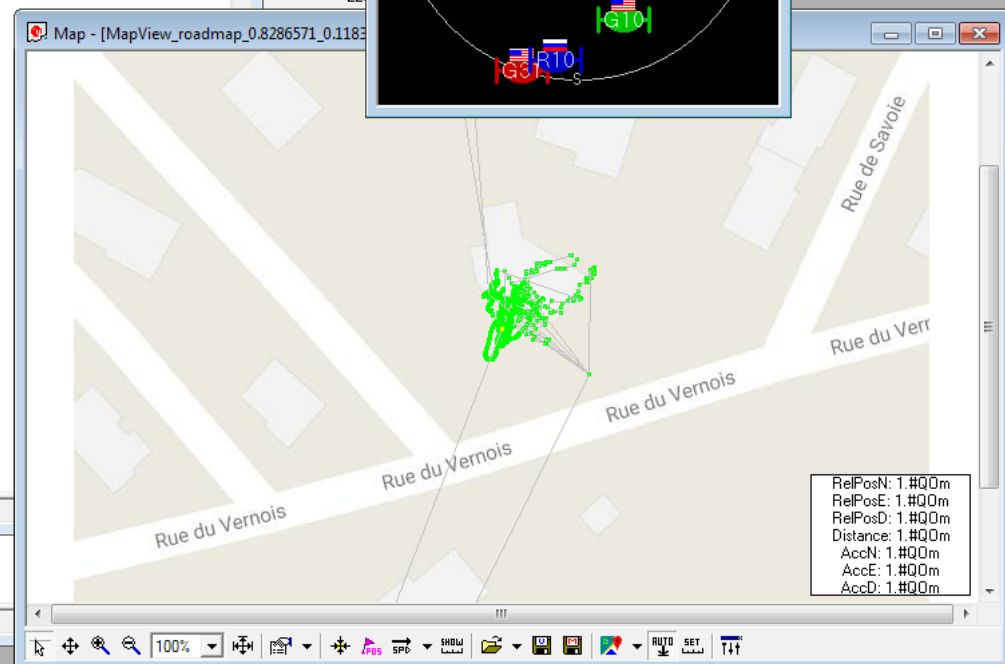
UBX - CFG (Config) - TMODE3 (Time Mode 3)

535 s

Packet Console

```
07:53:05 R -> NMEA GPGSV, Size 68, 'GNSS Satellites in view'
07:53:05 R -> NMEA GPGSV, Size 29, 'GNSS Satellites in view'
07:53:05 R -> NMEA GLGSV, Size 68, 'GNSS Satellites in view'
07:53:05 R -> NMEA GLGSV, Size 66, 'GNSS Satellites in view'
07:53:05 R -> NMEA GLGSV, Size 57, 'GNSS Satellites in view'
07:53:05 R -> NMEA GNGLL, Size 52, 'Geographic Position - Latitude/Longitude'
07:53:06 R -> NMEA GNRMC, Size 68, 'Recommended Minimum Specific GNSS Data'
07:53:06 R -> NMEA GNVTC, Size 35, 'Course Over Ground and Ground Speed'
07:53:06 R -> NMEA GNGGA, Size 82, 'Global Positioning System Fix Data'
07:53:06 R -> NMEA GNGSA, Size 58, 'GNSS DOP and Active Satellites'
07:53:06 R -> NMEA GPGSV, Size 68, 'GNSS Satellites in view'
07:53:06 R -> NMEA GPGSV, Size 68, 'GNSS Satellites in view'
07:53:06 R -> NMEA GPGSV, Size 29, 'GNSS Satellites in view'
07:53:06 R -> NMEA GLGSV, Size 68, 'GNSS Satellites in view'
07:53:06 R -> NMEA GLGSV, Size 66, 'GNSS Satellites in view'
07:53:06 R -> NMEA GLGSV, Size 57, 'GNSS Satellites in view'
07:53:06 R -> NMEA GNGLL, Size 52, 'Geographic Position - Latitude/Longitude'
07:53:07 R -> NMEA GNRMC, Size 68, 'Recommended Minimum Specific GNSS Data'
07:53:07 R -> NMEA GNVTC, Size 35, 'Course Over Ground and Ground Speed'
07:53:07 R -> NMEA GNGGA, Size 82, 'Global Positioning System Fix Data'
07:53:07 R -> NMEA GNGSA, Size 58, 'GNSS DOP and Active Satellites'
07:53:07 R -> NMEA GPGSV, Size 68, 'GNSS Satellites in view'
07:53:07 R -> NMEA GPGSV, Size 68, 'GNSS Satellites in view'
07:53:07 R -> NMEA GPGSV, Size 29, 'GNSS Satellites in view'
07:53:07 R -> NMEA GLGSV, Size 68, 'GNSS Satellites in view'
07:53:07 R -> NMEA GLGSV, Size 66, 'GNSS Satellites in view'
07:53:07 R -> NMEA GLGSV, Size 57, 'GNSS Satellites in view'
07:53:07 R -> NMEA GNGLL, Size 52, 'Geographic Position - Latitude/Longitude'
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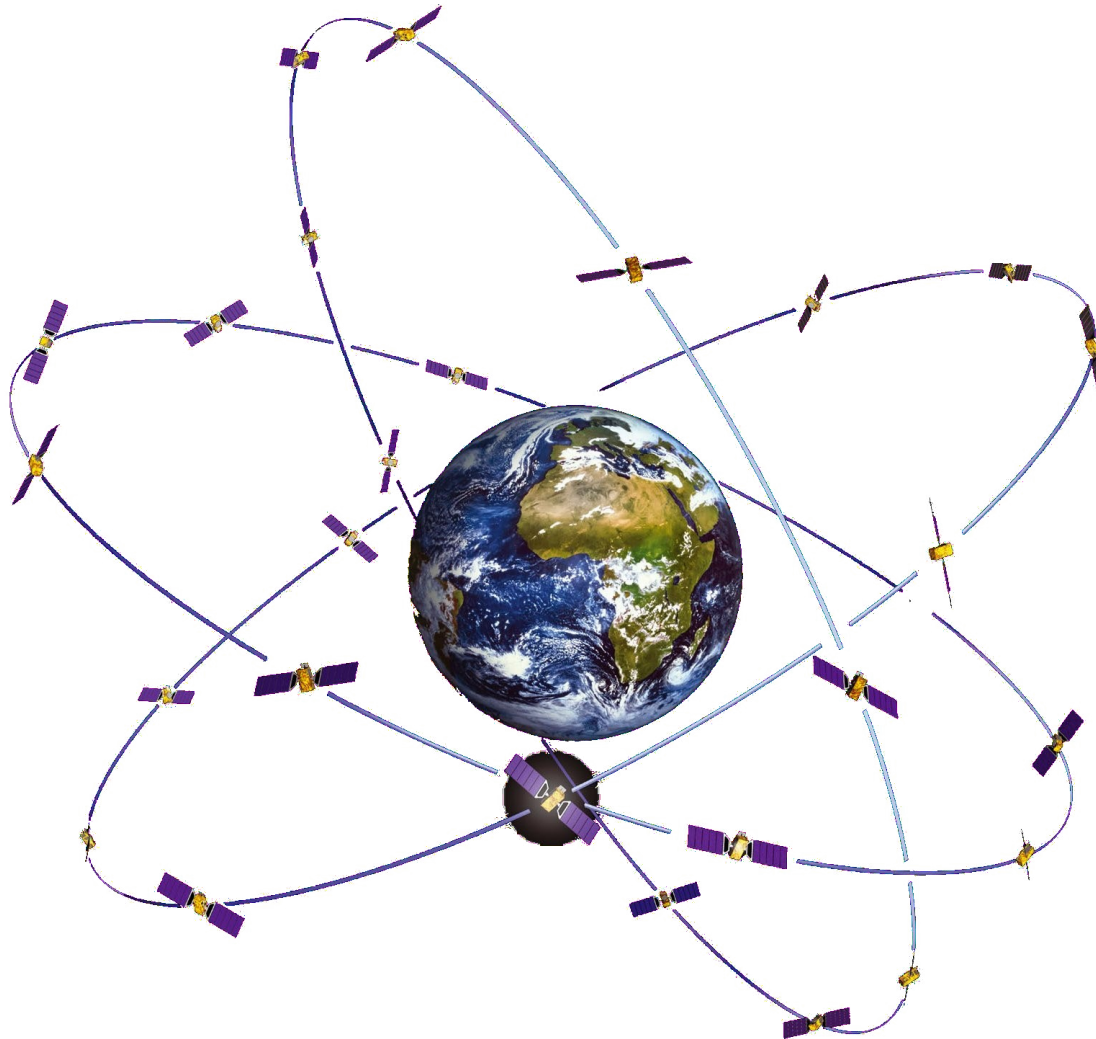
Rover Accuracy DGPS Float



Galileo: European Positioning System

- 09/2022 – 28 launched satellites
 - 24 active satellites
 - 2 satellites on a wrong orbit
 - 2 satellites out of order
- Coming next 10 additional satellites (launch until 2025)
- Spatial segment is fully active with 24 satellites
 - 8 satellites by orbit
 - 3 orbits
- Ground segment is active
- User segment
 - same hardware and same frequency (L1) than GPS
 - GPS and Galileo signals are compatible, but the software/firmware must be upgraded

Galileo: Satellite segment

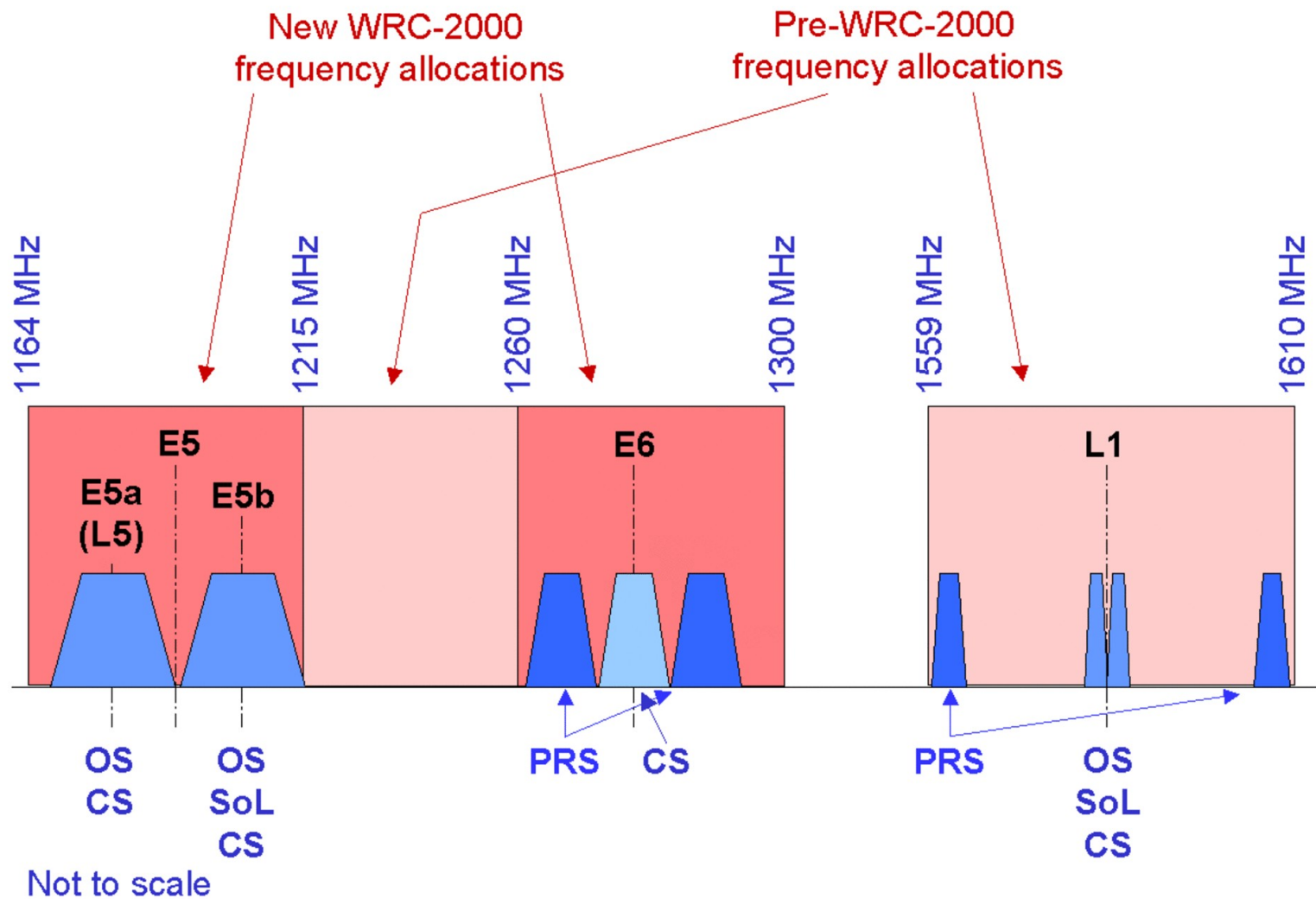


Galileo Services



- Open/Free Service (OS): simple positioning (mass market)
- High Accuracy Service (CS): encrypted signal, high and guaranteed accuracy,
- Safety of Life (SoL): open signal, integrity and authentication of signal
 - Using D-GNSS for accuracy
 - Using 2 or more coordinates calculus with different satellites for integrity
 - Using asymmetric algorithm to authenticate
- Public Regulated Service (PRS for Government): encrypted signal, integrity and continuity of signal
- Search and Rescue (SaR): Accurate coordinate and upload signal at 403MHz

Galileo Services

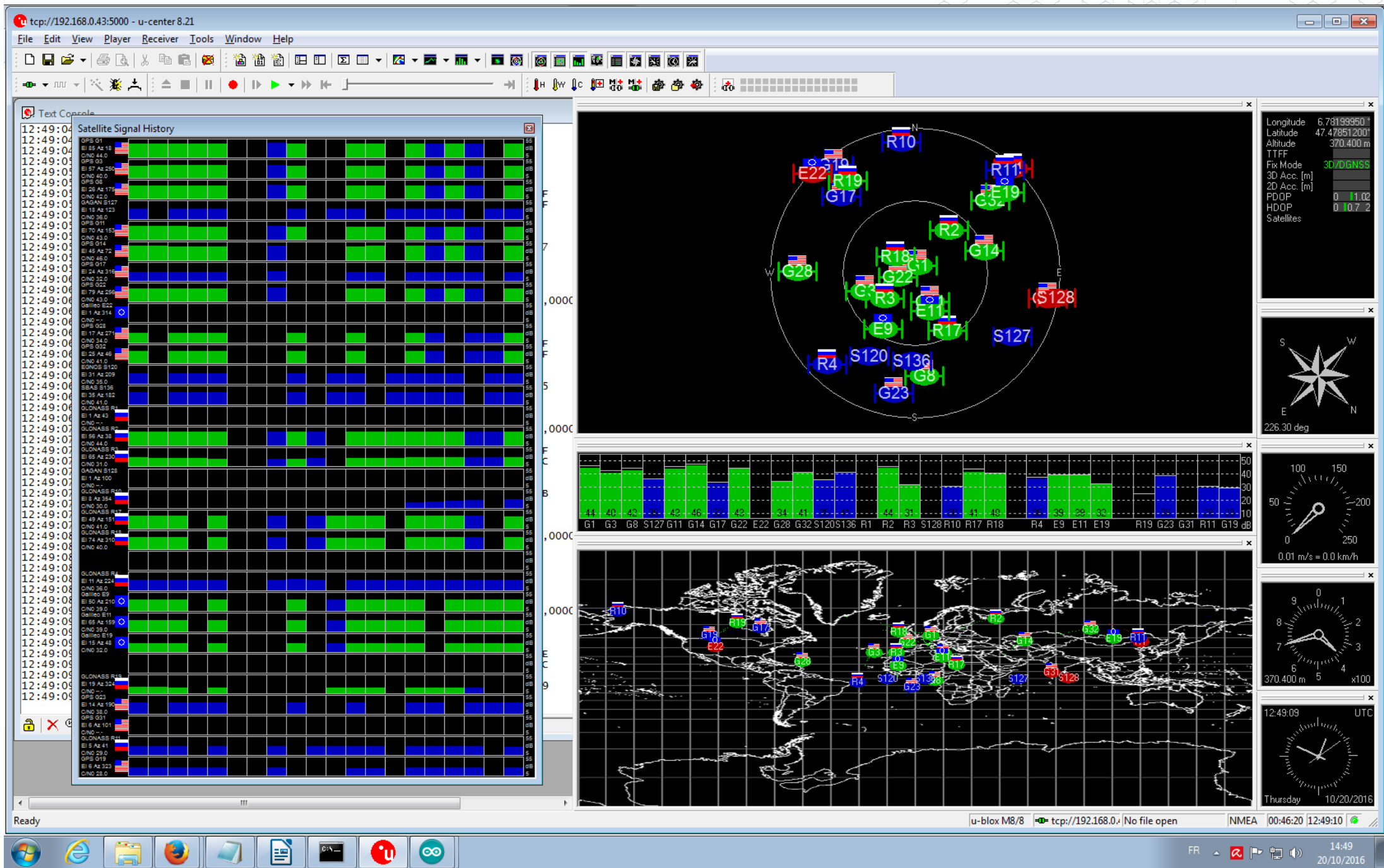


Galileo User Device: u-blox NEO-M8N

- One of the first user device including Galileo Satellites in its firmware (FW 3.01)
- Operates with the u-center software

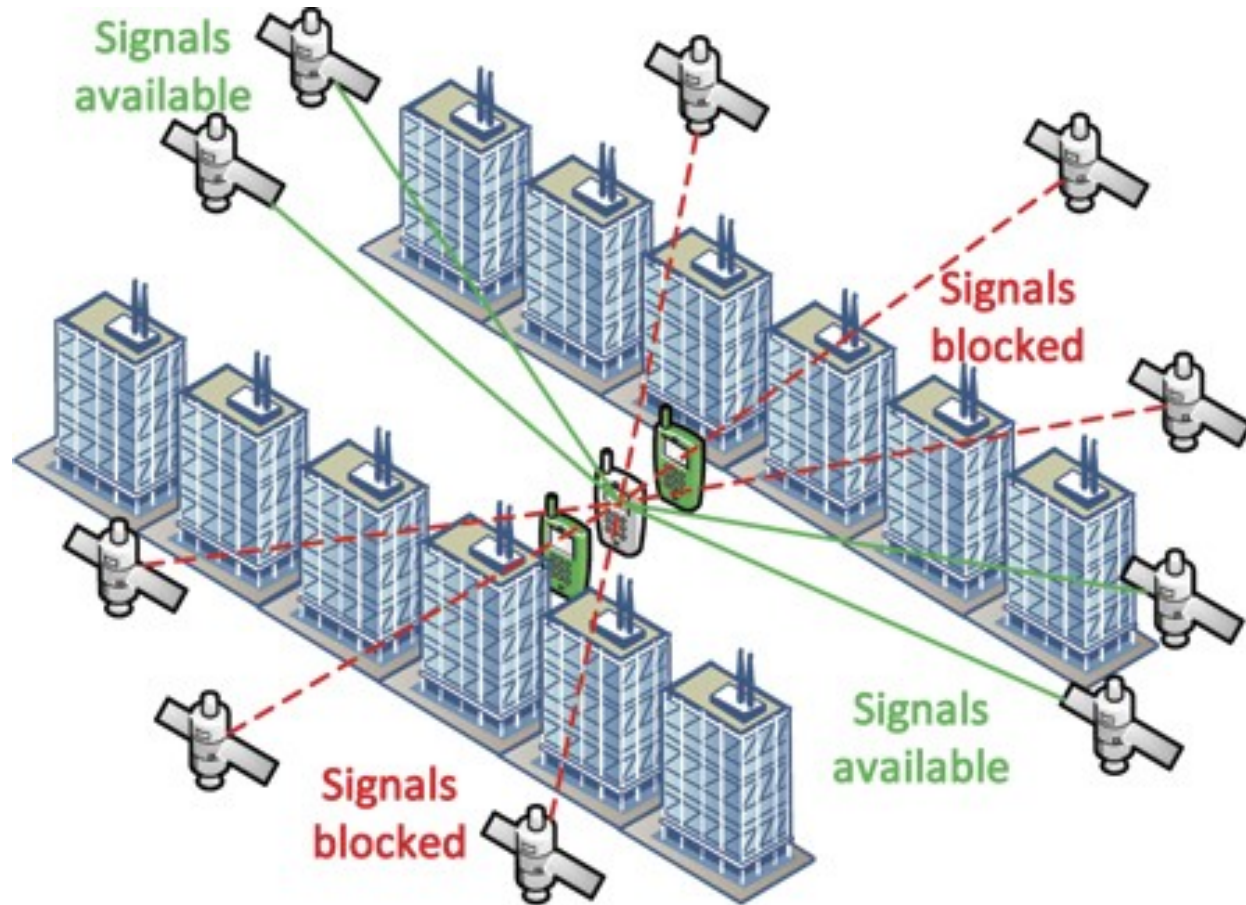


u-center software



Alignment effect in Urban Canyon

- In narrow streets or in front of high buildings, the visible sky is short and in only one axis
- Accordingly, the accuracy will be good in the street axis but will be very bad in the perpendicular axis of the street



Multi-constellation GNSS



- In order to have the best results, it is necessary to choose satellites as scattered as possible
- Instead of choosing at least the 4 best satellites in space from only one constellation such as GPS or Galileo, it is better to choose at least 4 best satellites in space from any constellation
- Today the 4 best satellites can be chosen among 30 navigation satellites

Indoor Application Context



- Commercial, Economic and Navigation uses.
 - Location of access-points inside the building is permitted and allows the access-points to be connected to power and network by wire for a long-term utilization. Coordinates of access-points are registered manually
- Fire and military operations
 - Access-points are deployed without wire and there is no time to register manually the coordinates for a short-term utilization
 - Location of any mobile terminal inside the building is useful
 - In case of fire, location of access-points can be done outside and inside the building.
 - In case of military operation, location of access-points must be installed only outside the building.
- Security in building
 - Needs to locate all the mobile terminals inside the building from registered and unregistered terminals

Properties for Indoor Positioning

Accuracy for fixed and mobile situations

Availability of the system

Integrity of calculated coordinates

Security and **confidentiality** of the data exchange

Adaptability of the system to the environment building

Coverage and density of access-points to deploy

Categories of Indoor Systems



- Radio Systems
 - Signal Strength Systems
 - Time of Flight Systems
- Inertial Systems
- Ultrasound Systems
- Camera Systems
 - Artificial Marker
 - Natural Marker
 - SLAM: Simultaneous Location and Mapping

Homogeneous Area



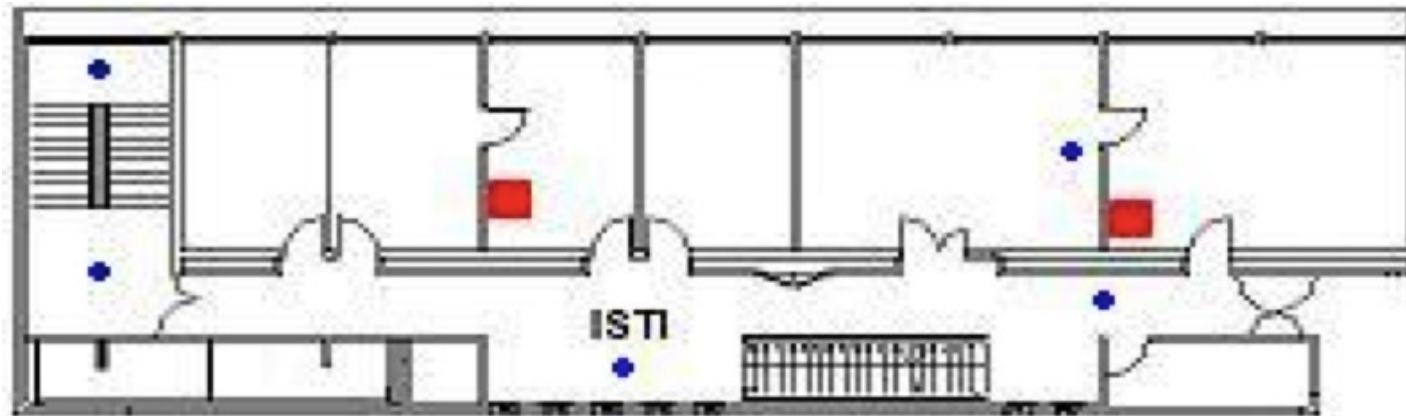
- Radio signal strengths can be defined
- Trilateration calculus is available
- Friis Equation mostly correct
- Good accuracy in such a context, even without any feedback function

Heterogeneous Area

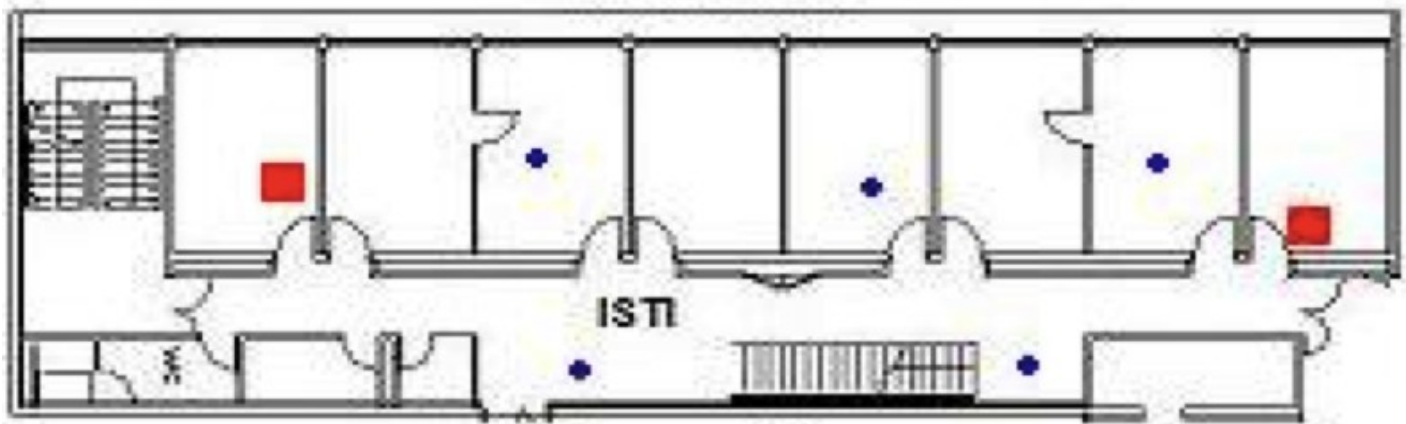


- Signal strength difficult to determine
- Very high rate of multipath (fading and fast fading)
- Variation of context
 - short term
 - People which are moving
 - Opening and closing of windows and doors
 - Long term
 - Adding and removing of partition inside the area
 - Changing ways of electricity or water
 - Moving of furniture
- Need for an adapted propagation
- Weak accuracy without feedback function or a priori measurements (benchmark)

Heterogeneous area example



First Floor

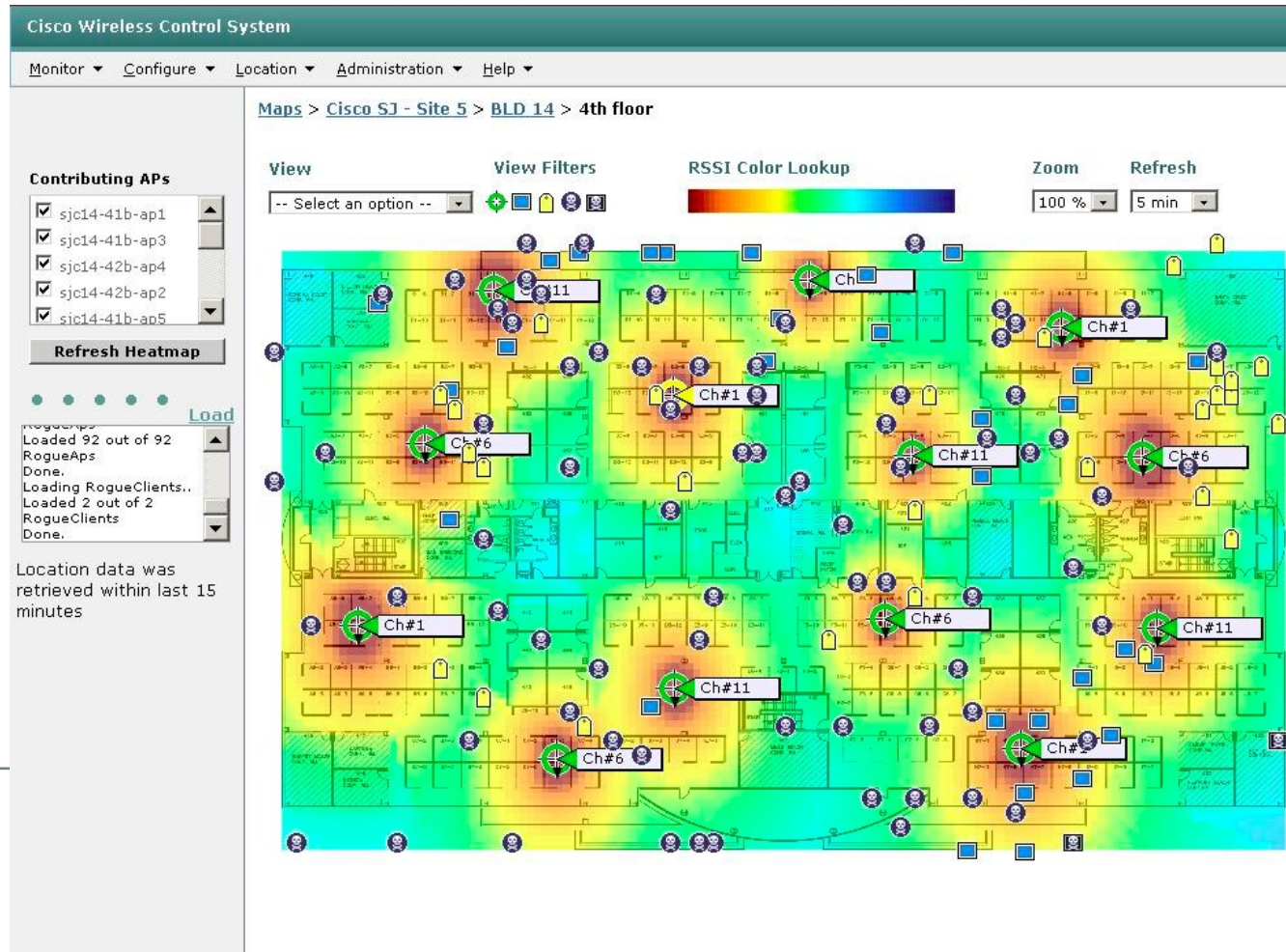


Ground Floor

Radio Solutions for Indoor Positioning

- Need for a level of a received signal strength
 - Use of a standard propagation formula
 - Collecting a received signal strength map (fingerprinting)

Very difficult to build the fingerprint without going inside the building before

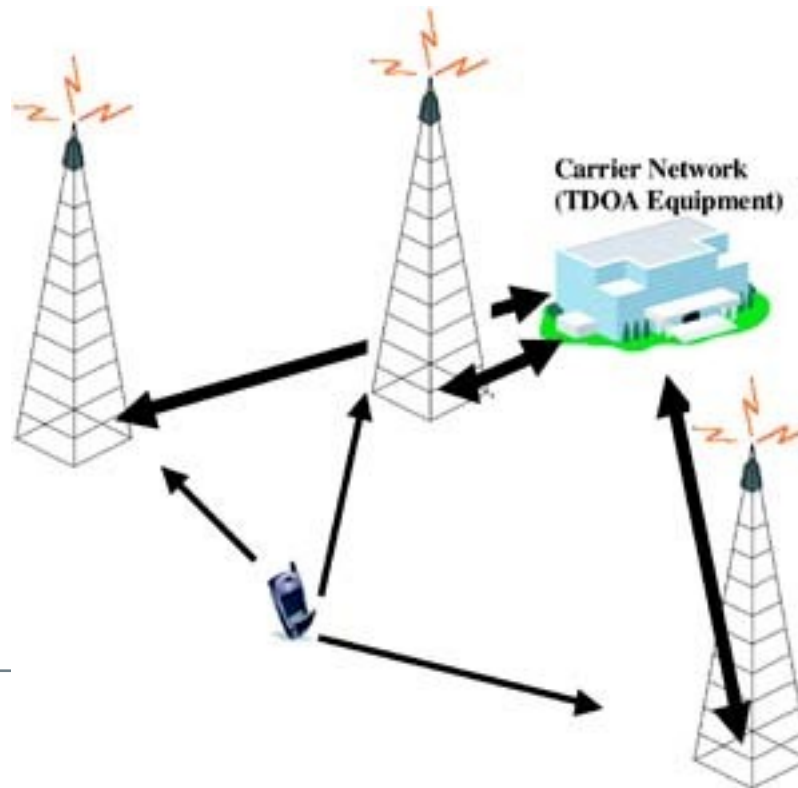


Indoor Positioning solutions



- Time of Flight Measurements of a radio signal (GSM or UWB)
 - ToF : Time of Flight
 - ToA : Time of Arrival
 - TDoA : Time of Differential of Arrival

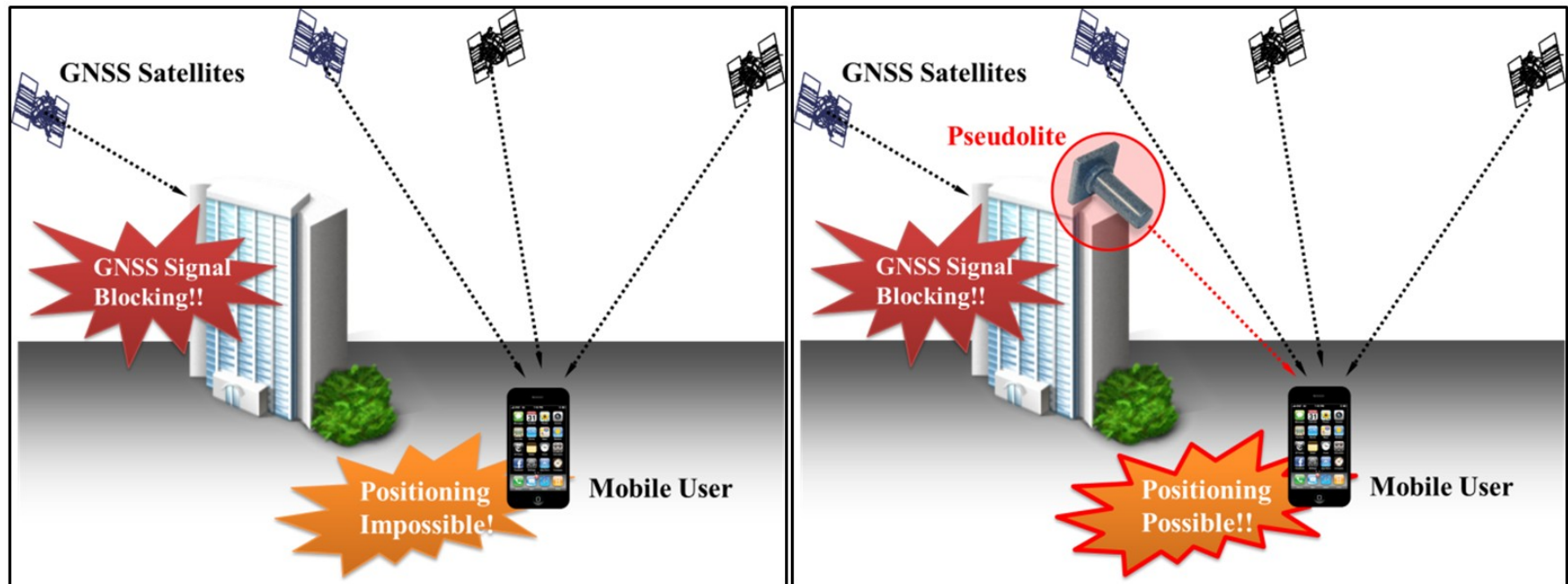
Hard to calculate
correct coordinates in a
multipath environment



- Base transceiver stations (BTSs) receive signal from handset
- TDOA equipment
 - Measures the difference in the time the signals arrive at the BTS
 - Translates this data into latitude and longitude

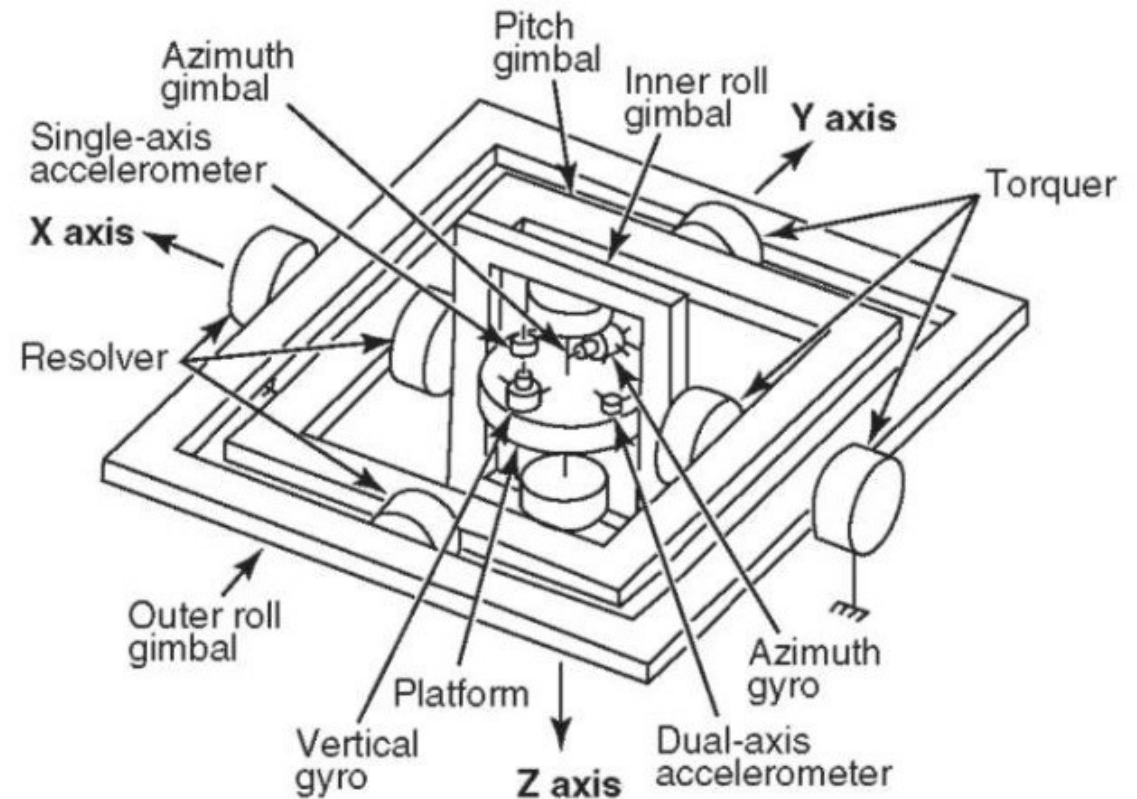
Indoor Positioning Solutions

- Pseudolite: by adding transmitters inside the building using the GNSS frequency in order to substitute the satellite signals



Indoor Positioning Solutions

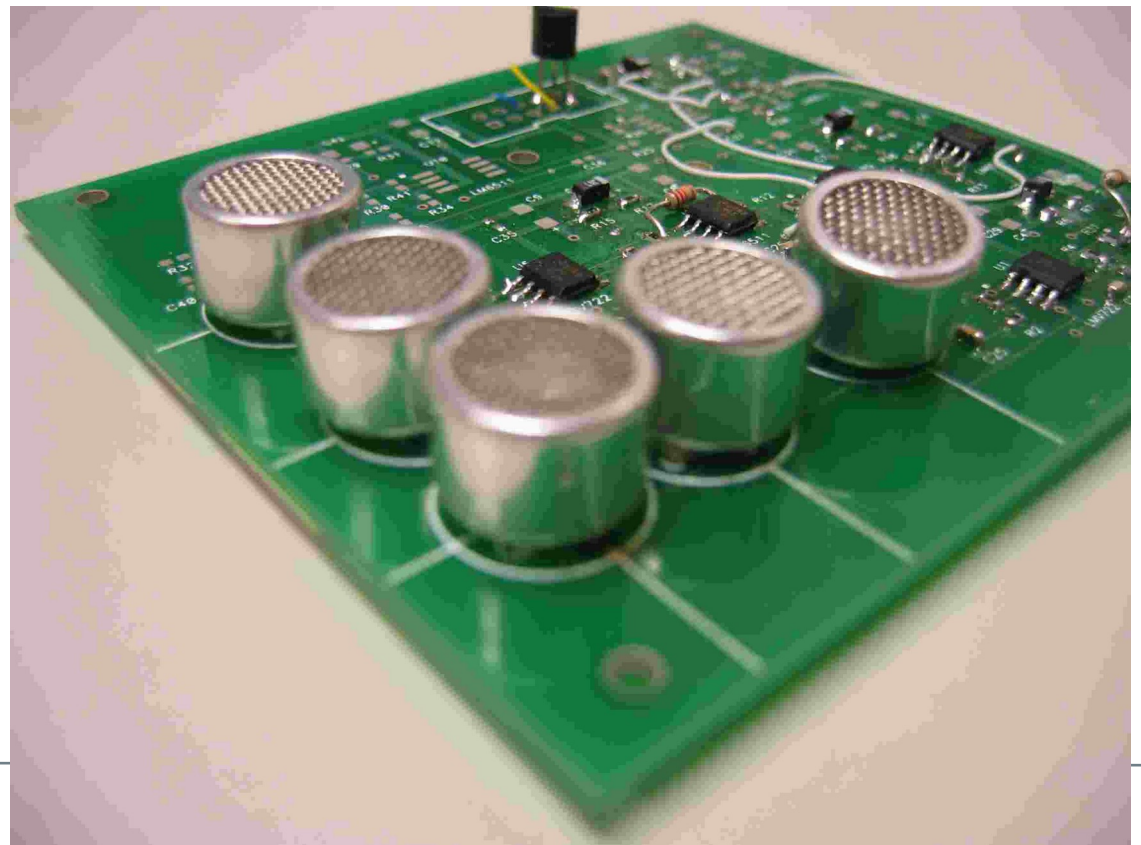
- IMU : Inertial Measurement Unit – works in a relative context
- Difficult to follow sudden movements
- Cumulative error over time



Indoor Positioning Solution



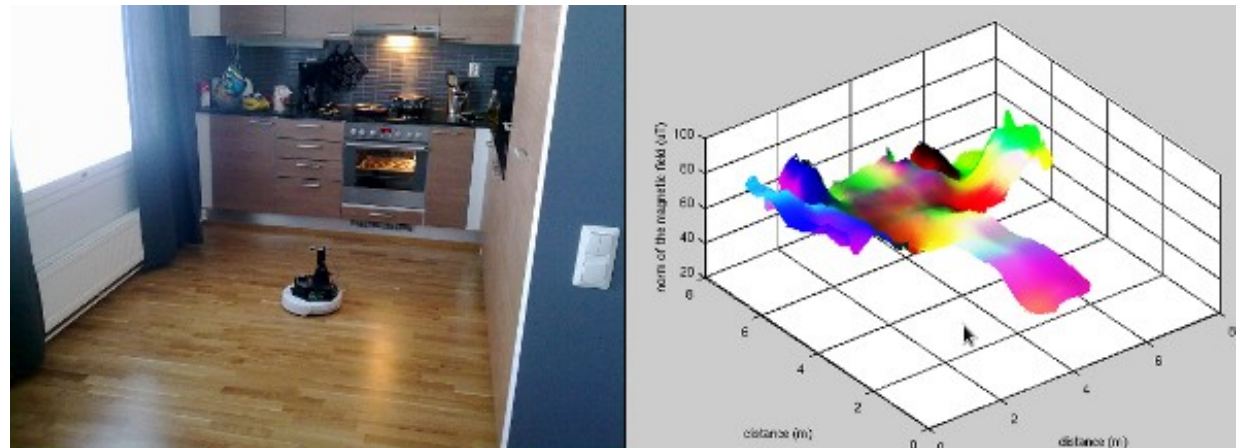
- Ultrasound measurements
- works well inside a room, but difficult to cross walls and obstacles



Indoor Positioning Solutions



- Recognition and identification of areas by using cameras – SLAM
– Simultaneous Localization and Mapping



Discovery room by room

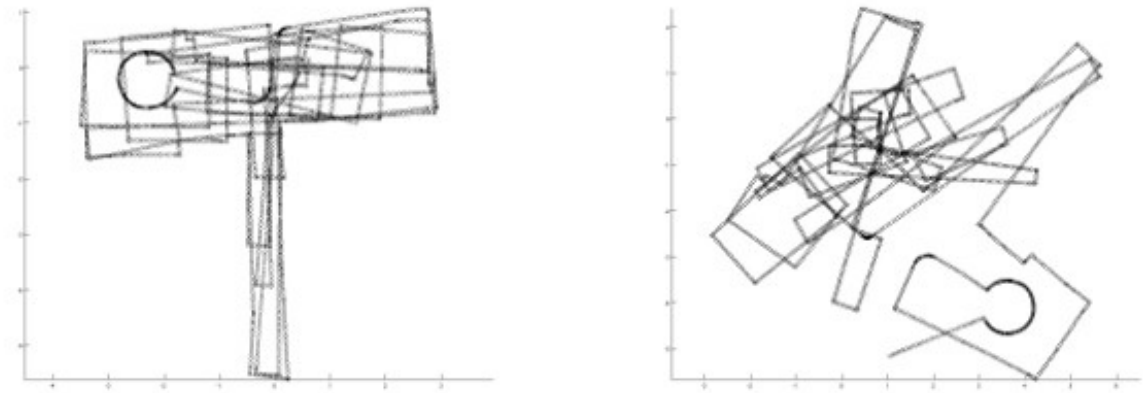
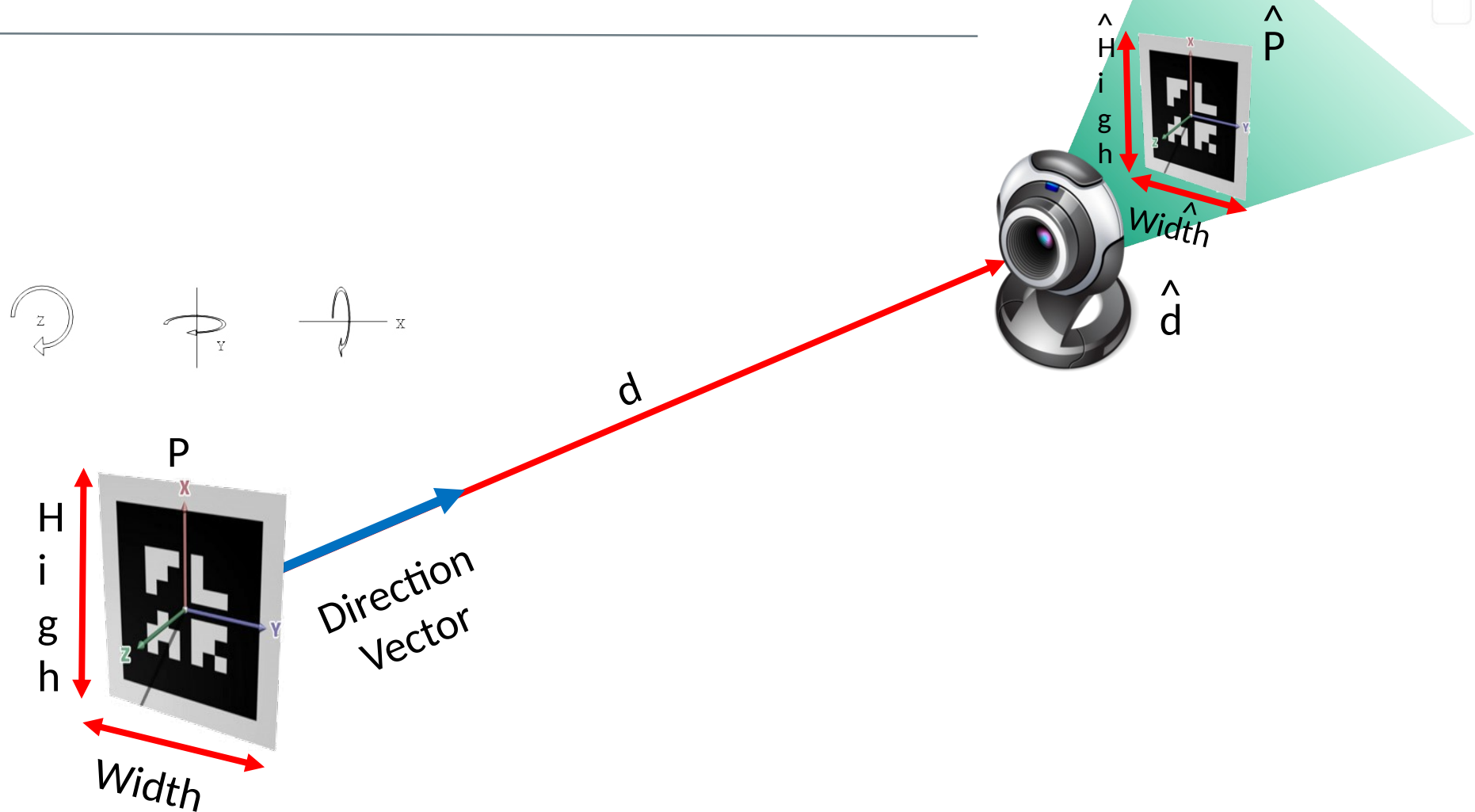


Image processing by finding artificial or natural markers



Synthesis of Indoor Positioning systems

System	Accuracy	Indoor / Outdoor	Conditions
GSM	hm	Outdoor	Open Environment Good GSM coverage (Number of antennas)
GPS	dam - m	Outdoor	Open Environment Good geomtry of satellites
Wi-Fi	m	Indoor/ Outdoor	Select non-noisy channel, minimize objects
DGPS/RTK	cm	Outdoor	Open Environment
Camera	cm	Indoor / Outdoor	Good lightening condition (>80 lux) Low speed movement (to avoid blurred) Good color variation in the environment
Inertial/compass	cm to m	Indoor/ Outdoor	Avoid electromagnetic interference

History of Wi-Fi Indoor Positioning

~2000 – First Wi-Fi Access-Points with closed system

2004 – First Wi-Fi Access-Point with an open system

Linux/openWRT

Researches activities in Indoor Positioning Area

1995/2005 – Indoor Positioning systems from the user device

2005/2015 – Indoor Positioning – Infrastructure-centric



Data receiver solution



Because Wi-Fi signals are bidirectional, both the terminal and the access-points can be used as a receiver to measure the signal strength.

A contrario, GNSS signals are unidirectional and only the terminal can receive the signal from satellites.

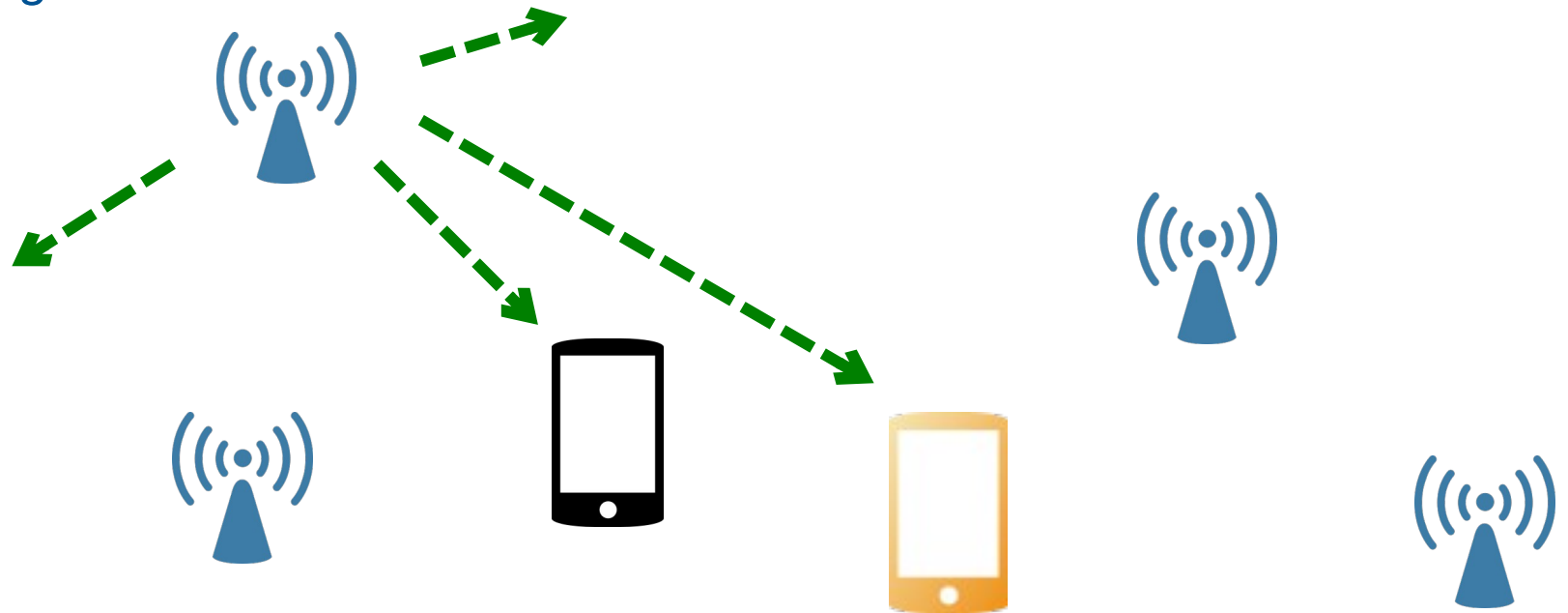
With Wi-Fi signals, there is three ways to develop a location service:

- Mobile-centric solutions use the receiver capabilities of the terminal to collect at least three access-point signals
- Infrastructure-centric solutions use the receiver capabilities of at least three access-points to measure the same signal sent by a terminal.
- both solutions can be used simultaneously in order to improve the integrity

The mobile-centric solution

Mobile-centric solutions are quite difficult to develop for each mobile Operating System such as iOS, Android, GNU/Linux or other proprietary systems. Collecting the received strength signal is a low function not always delivered by the OS.

Advantage: one signal transmits by one access-point can be received by all the terminals in coverage

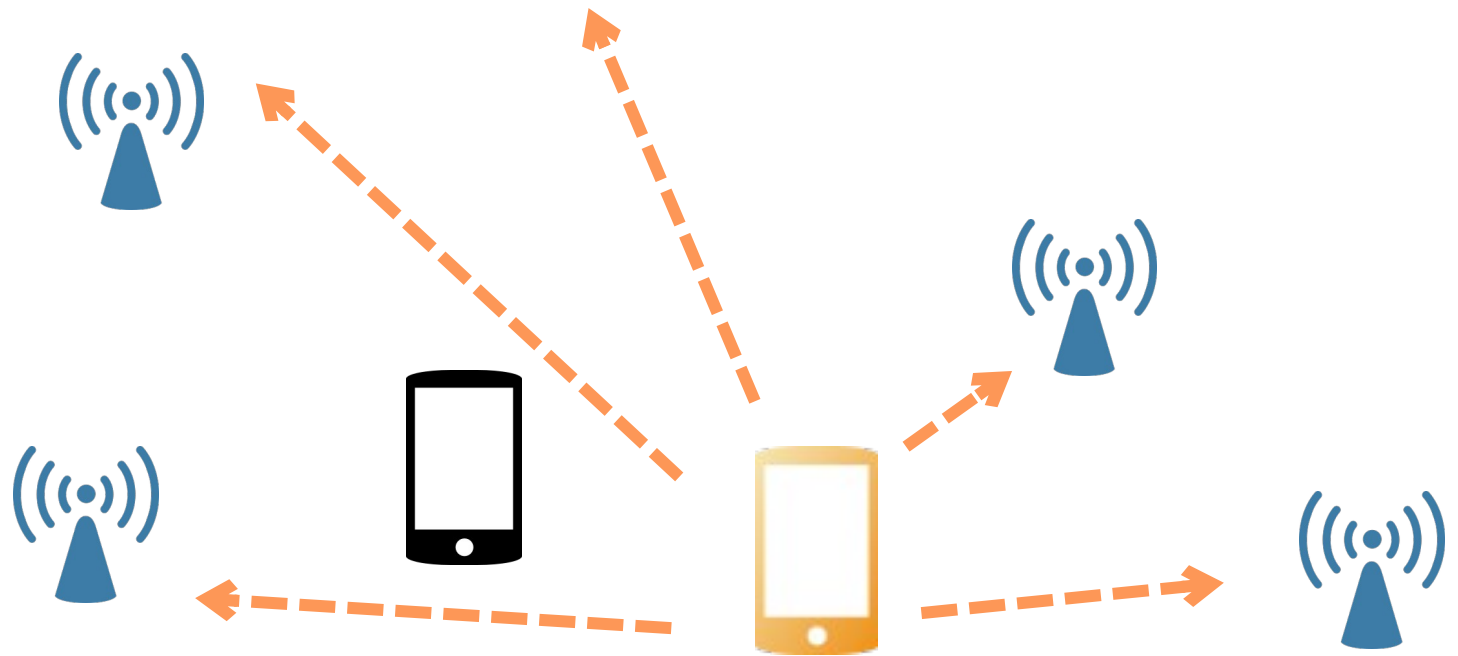


The Infrastructure-centric solution

Infrastructure-centric solutions use the receiver capabilities of at least three access-points to measure the same signal sent by a terminal. Data must be aggregated in the network infrastructure to compute the position.

Advantage: only one packet sent can conduct to calculate a position

Drawback: each mobile must sent packets to be localized



The Infrastructure-centric solution

- Because, each access-point must deliver the received signal strength and sent the value to an aggregator/location service, it is necessary to add some specific algorithm:
 - Programmable access-point: WRT-54G or Foner
 - Linux embedded system
e.g. Raspberry PI
 - Arduino programming without wire



Raspberry PI



Linksys WRT54G



Esp-8266

Context for using Wi-Fi signal strength

- Define an accurate propagation model in a heterogeneous environment
 - Difficulties to calculate an accurate pseudo-range distance between a transmitter and a receiver
- Define a manual mapping of the measured receiving signal strengths called Fingerprinting
 - Initial phase needs to store the signal strength all over the building
 - Registered values will become obsolete over time because of modification in building, people inside the building, temperature...
 - Building an accurate fingerprint takes time

Approach to improve fingerprinting

- Our solution is to define a fully automated phase to calibrate the positioning system. The solution is to send signals between access-points
 - Access-points are located at known coordinates
 - Environment quality is estimated specifically for each link between two access-points
 - Calibration step can be ran permanently or regularly in order to modify the environment quality to the current condition
 - Integrity can be evaluated using more than 3 access-points

Infrastructure deployment strategy

- Long-term use
 - Access-points are located inside the building
 - Access-points are connected with wires to building plugs
 - Access-points coordinates are manually configured
- Short-term use
 - Access-points are located outside and inside the building
 - Access-points are self-powered using battery without wire
 - Access-points must calculate automatically their coordinates using Line of Sight technology such as UWB, Differential GNSS, IR light or ultra-sound

Design of the OwlPS Indoor system

Open Wireless Positioning System (University of Franche-Comté)

Hardware

Wi-Fi Access-Point – Fonera – 200MHz MIPS Processor – 8MB flash – dedicated Wi-Fi Atheros chipset

Indoor Positioning System

Capture of Wi-Fi traffic on the Access-Point

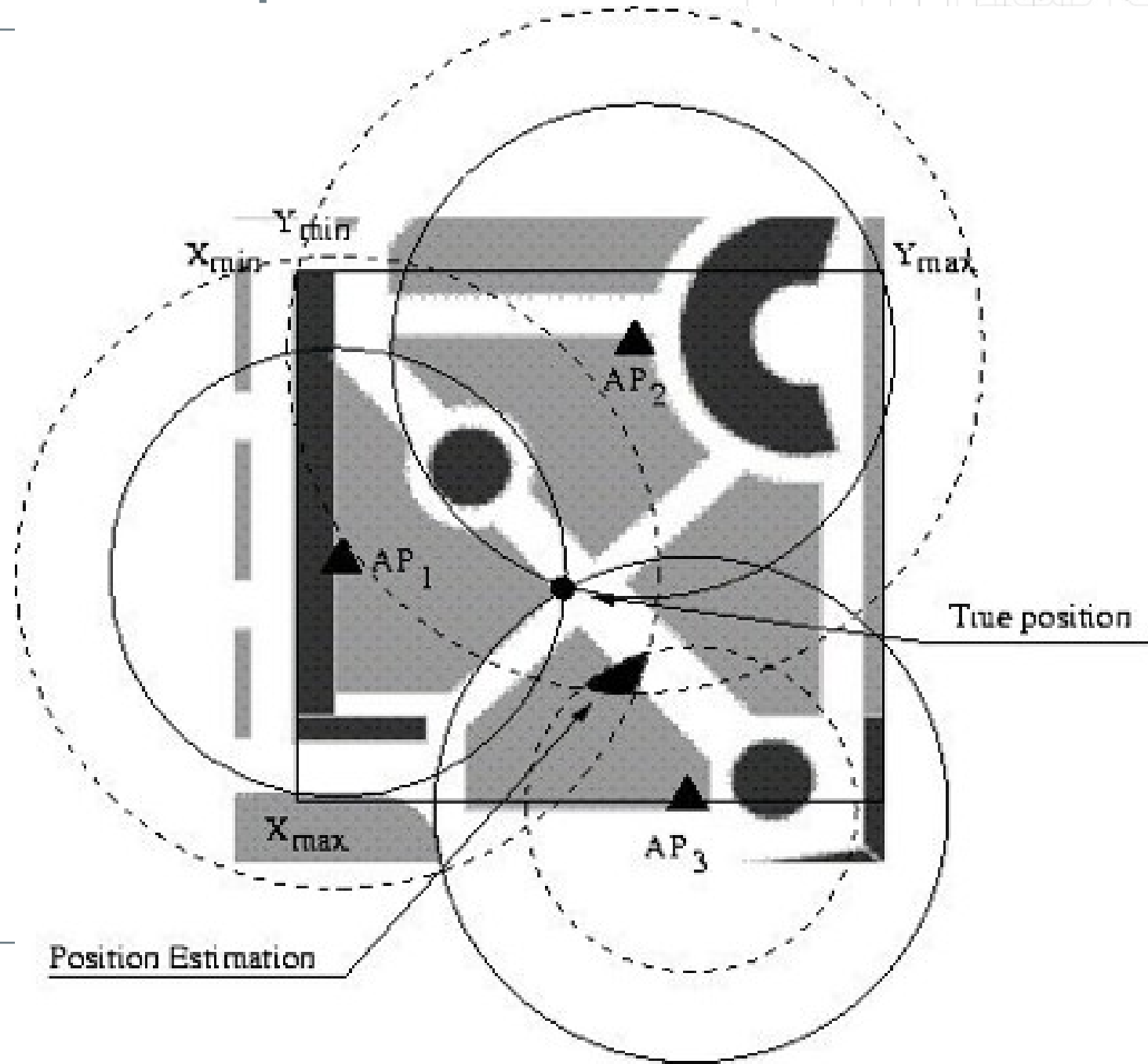
Measure of Signal Strength from each received packet (Radio TAP library)

Data Transmission to a aggregation Server

Position Calculus using trilateration

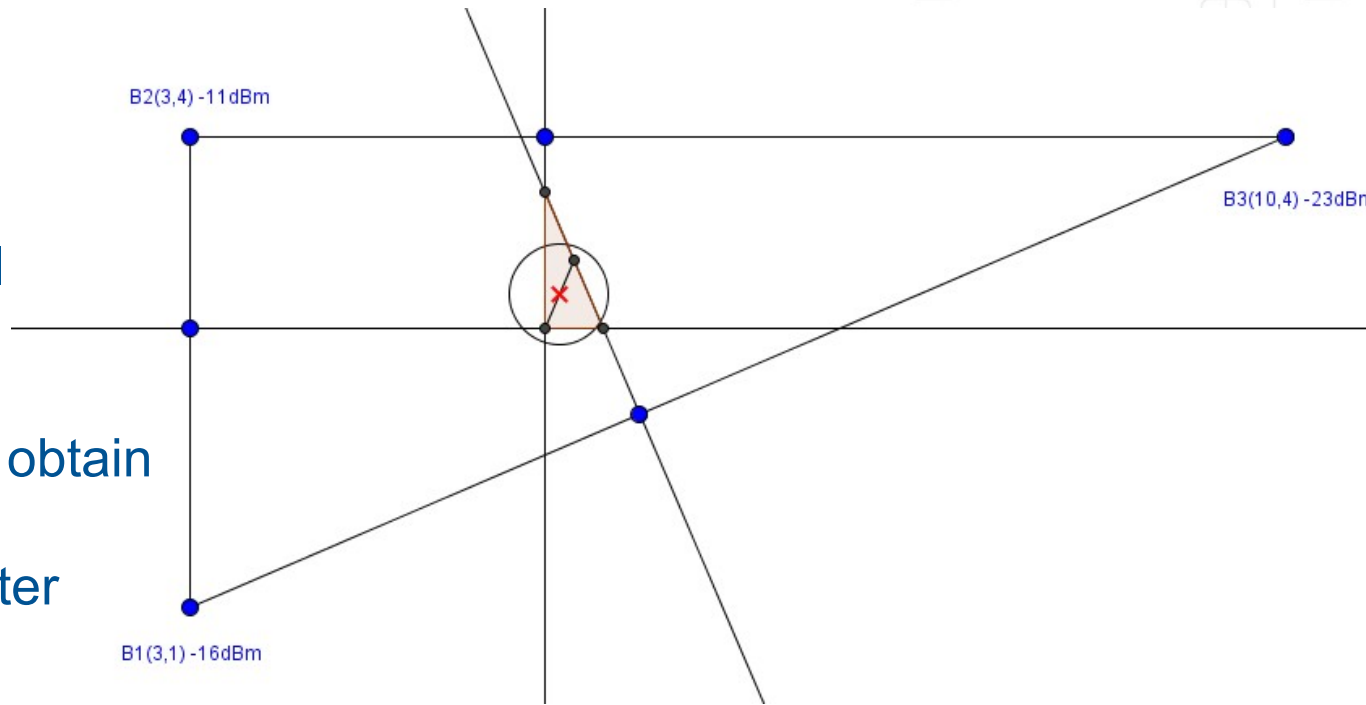


Trilateration Principle



Weighted perpendicular bisector lines

- Instead of crossing 3 circles, we weighted a perpendicular bisector line in order to take into account the received signal strength on two different APs.
- Then we intersect the 3 signed weighted perpendicular lines to obtain a triangle
- Finally, we calculate the perimeter of the triangle as a dilution of precision (uncertainty).
- We iterate this calculus with other combinations of 3 APs in coverage
- At the end, we can compare, the proximity of estimated positions according to the smallest triangle perimeters.
- With this method, it is easier to guarantee the position and its integrity



List of Usable Wireless Networks

- Mobile Cellular Network (GSM, UMTS, LTE, ...)
- LPWAN (Low Power Wide Area Network, e.g. LoRaWan/Figfox)
- Wi-Fi Network (2.4GHz and 5GHz)
- Bluetooth Network (short range)
- RFID (Radio Frequency ID)
- Ultra Wide Band (UWB)

Mobile cellular Network



- Measurements done using
 - Integrated functionalities inside the base station
 - By sending a request for a reception report by the infrastructure to a user device-smartphone

Wi-Fi Network



- Measurements done using
 - RSSI : Received Signal Strength Indicator
 - Normalized for the connected Access-Point
 - Linked to each received packet (hard to access)
 - There is no dedicated standard for Positioning System, but there are possibilities to access to a few low level information

Bluetooth Network



- Measurements done using
 - RSSI : Received Signal Strength Indicator
 - Readable only when an association has been done
 - There is no standard according to the Positioning System
 - Short coverage

Ultra Wide Band (UWB) communication

- Radio pulse on a wide band (500MHz)
- Civil utilization authorized since 2002
- Main Norm: 802.15.4a
- Coverage distance around 10/15m
- Accuracy 5 to 10cm in case of LoS

Classification of Positioning Model

- « Theoretical » Propagation Model
- Model using Signal Strength Measurements
- Geomteric Model
- Hybrid Model

« Theoretical » Propagation Model

- Friis Equation

$$\frac{P_R}{P_T} = G_R G_T \left(\frac{l}{4 \pi d} \right)^2$$

- Hardly usable in an heterogenous environment

Modeling using Signal Strength Measurements

- Building of Signal Strength Map as an Input (Calibration or Fingerprinting)
- Storing results on a database
 - Coordinates of each location
 - Signal Strength of each received signal
 - Device orientation
- Accuracy between 2 to 5 meters

Geometric Model



- Map of the building
- Ray tracing and simulation of all the physical phenomena in the area
- Difficult to integrate short term variation as opening doors or people movements
- In good condition, accuracy around 1 meter
- Time to model the building can be quite long

Integration of the measurement function

- In a bidirectional network, the measurement function can be included
 - In the mobile device, centralized solution
 - In the network infrastructure (Access-Point), distributed solution

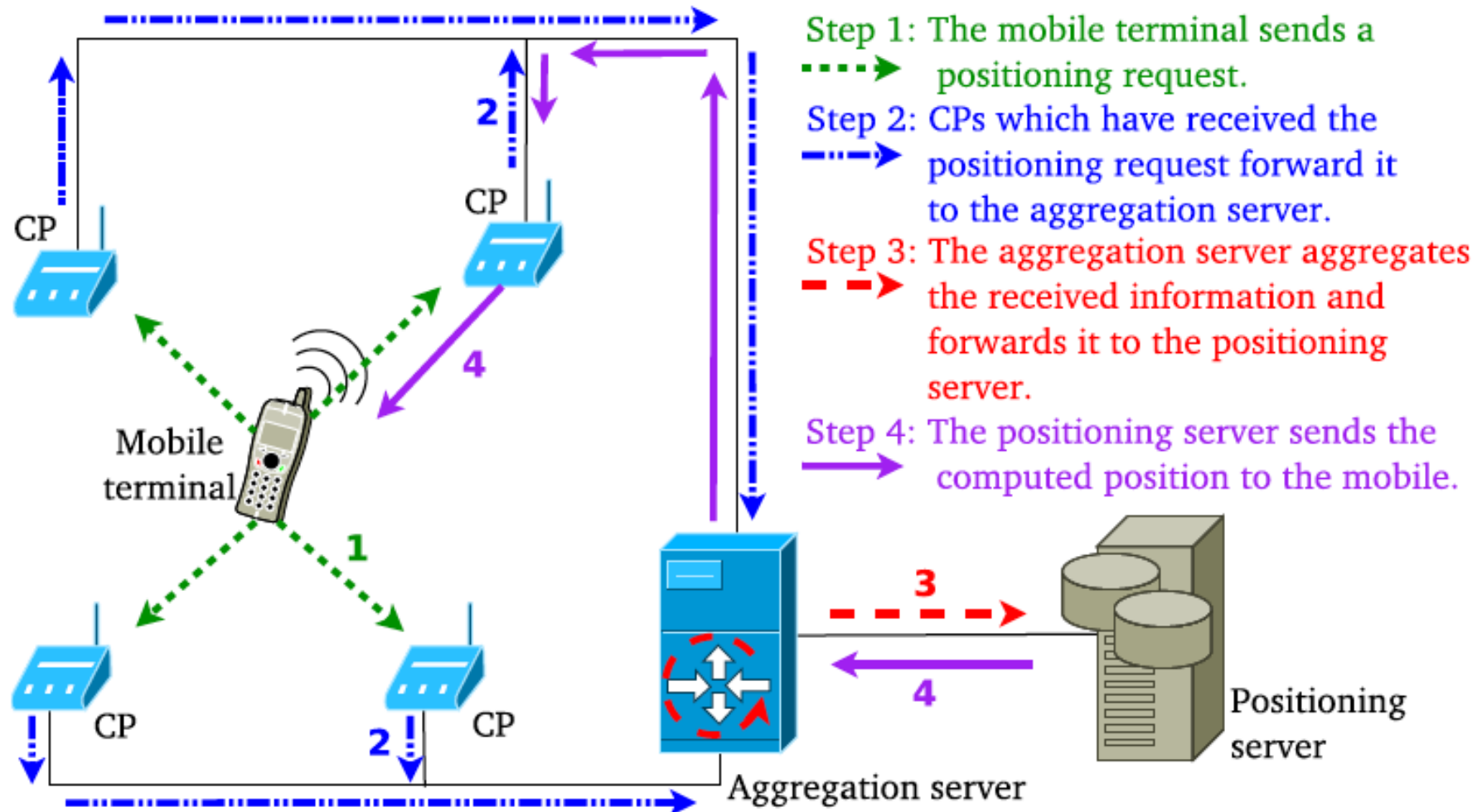
Integration inside the mobile device

- The mobile device must do a signal strength map or reading a result
- Mainly, the device send its measurements to the infrastructure
 - Using a normalized report (GSM)
 - Using e dedicated piece of code (Wi-Fi)
- Target applications
 - Services for the device
 - Monitoring of authorized devices

Integration inside the network infrastructure

- Base stations or Access-Points realized measurements concerning one device
 - It is necessary to centralized the measurements in a service in order to calculate the position
 - Dedicated software inside the Access-Points
 - No software in the mobile device
- Target Applications
 - Services for the device
 - Monitoring, security in the building and intrusion detection

Our Platform OwlPS



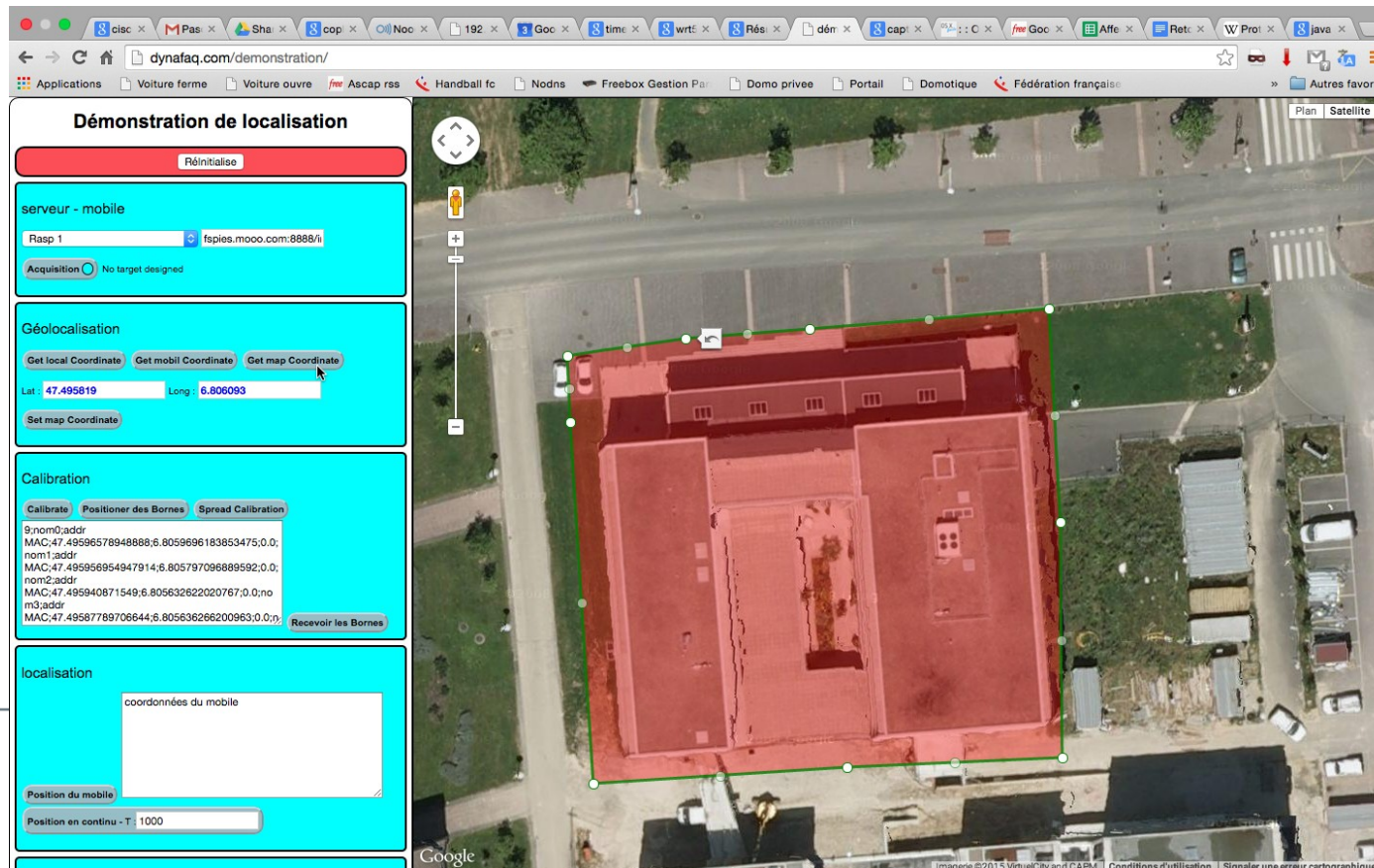
Advantages of OWLPS (1/2)



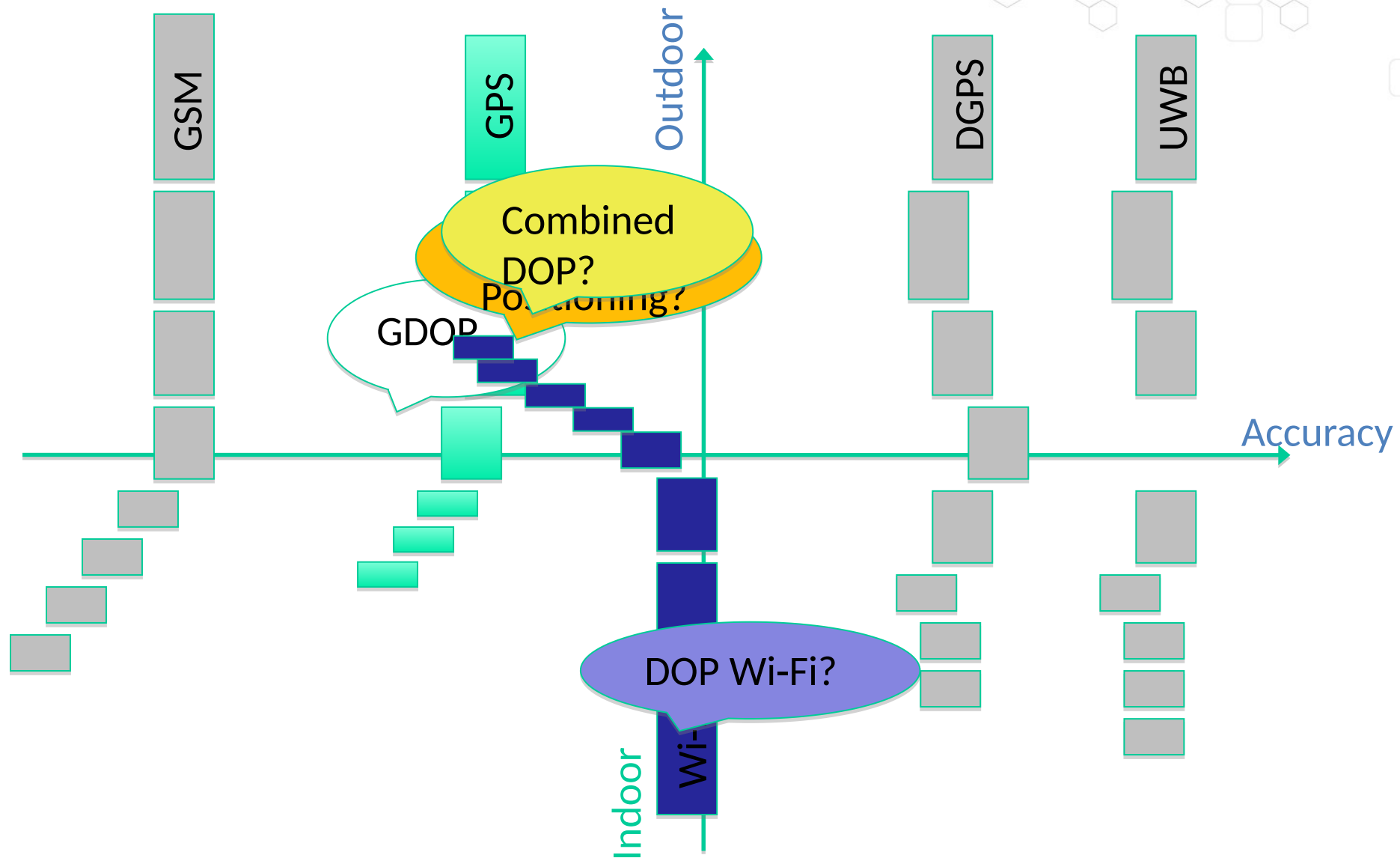
- Calculus in the network infrastructure
 - Accurate positioning of our devices using periodic signals – if necessary
 - Positioning of other devices (unknown or hostile) – without traffic injection
 - Signal Strength Variation observation between Wi-Fi APs allowing to identify indoor movements – decrease of signal: people on the trajectory or increase of signal: opening doors

Advantages de la plate-forme OWLPS (2/2)

- Initial location of AP inside the building
- Allowing deployment of APs outside near from the building without going inside the building.



Comparison of Positioning systems

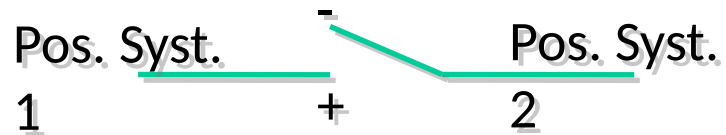


Continuity of the Positioning Service

- No universal system allows to cover all the situations
- Need of 2 or more positioning systems
 - Increase of the application domain
 - Quality evaluation of the calculated position
 - Management of changes from one system to another
- Switching solution
- Combining solution

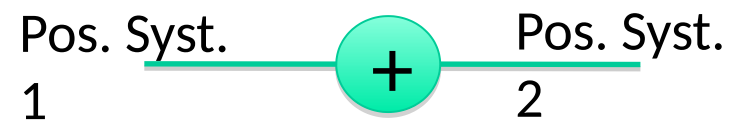
Continuity of the Positioning Service

Positioning using switching



- Few Positioning Systems
- Only one System at a time

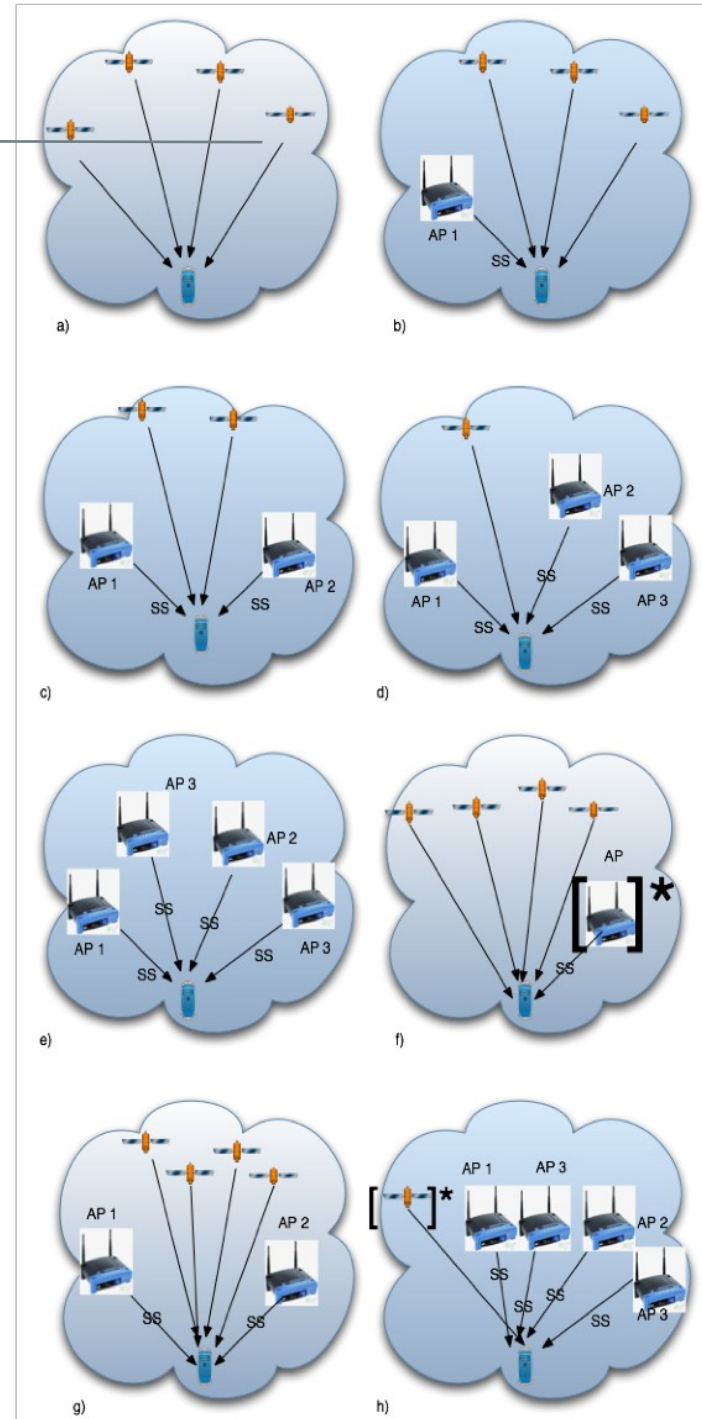
Positioning using combination



- Few Positioning System
- Use of transmitters from available positioning Systems

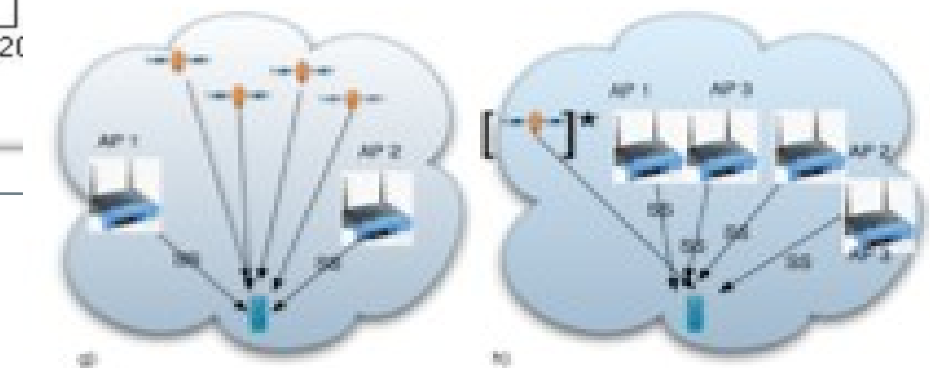
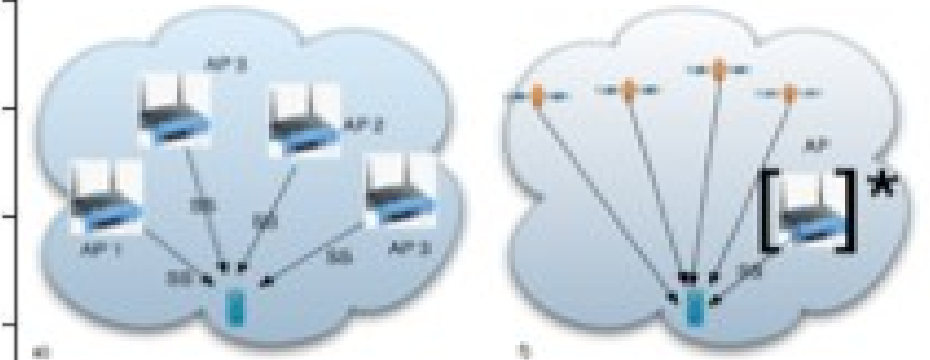
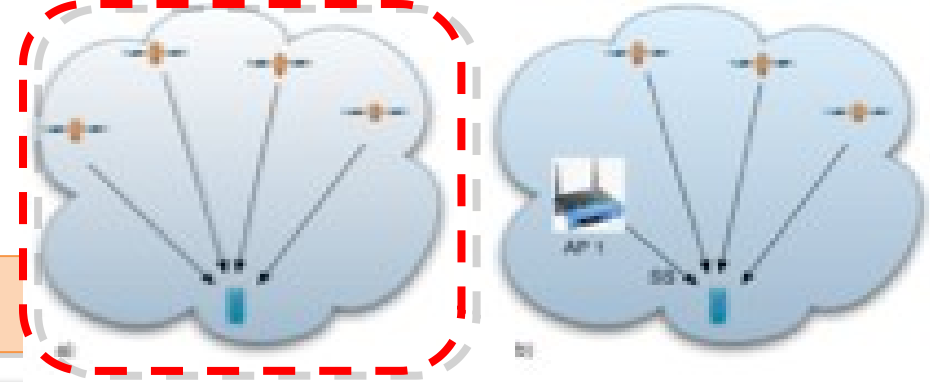
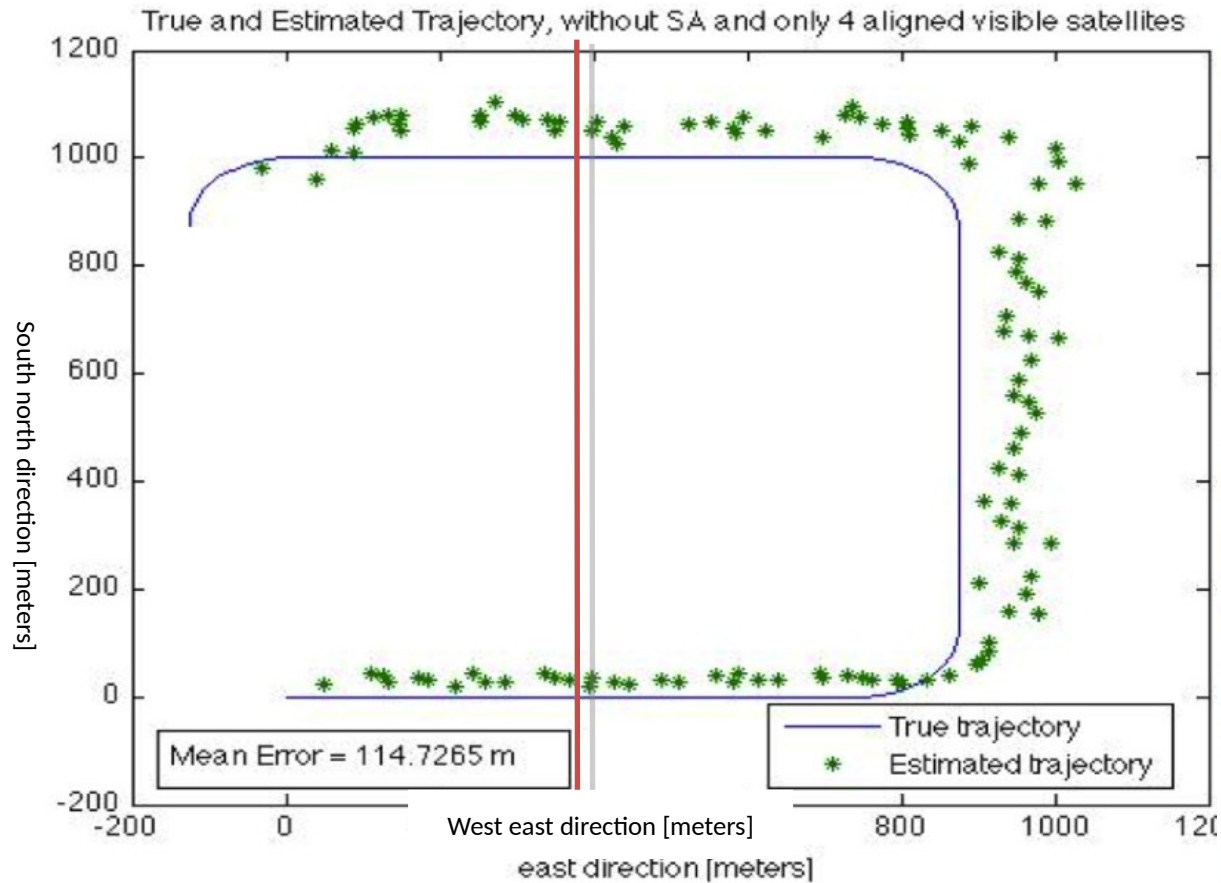
Principle of a Combined system

Various Repositories Association



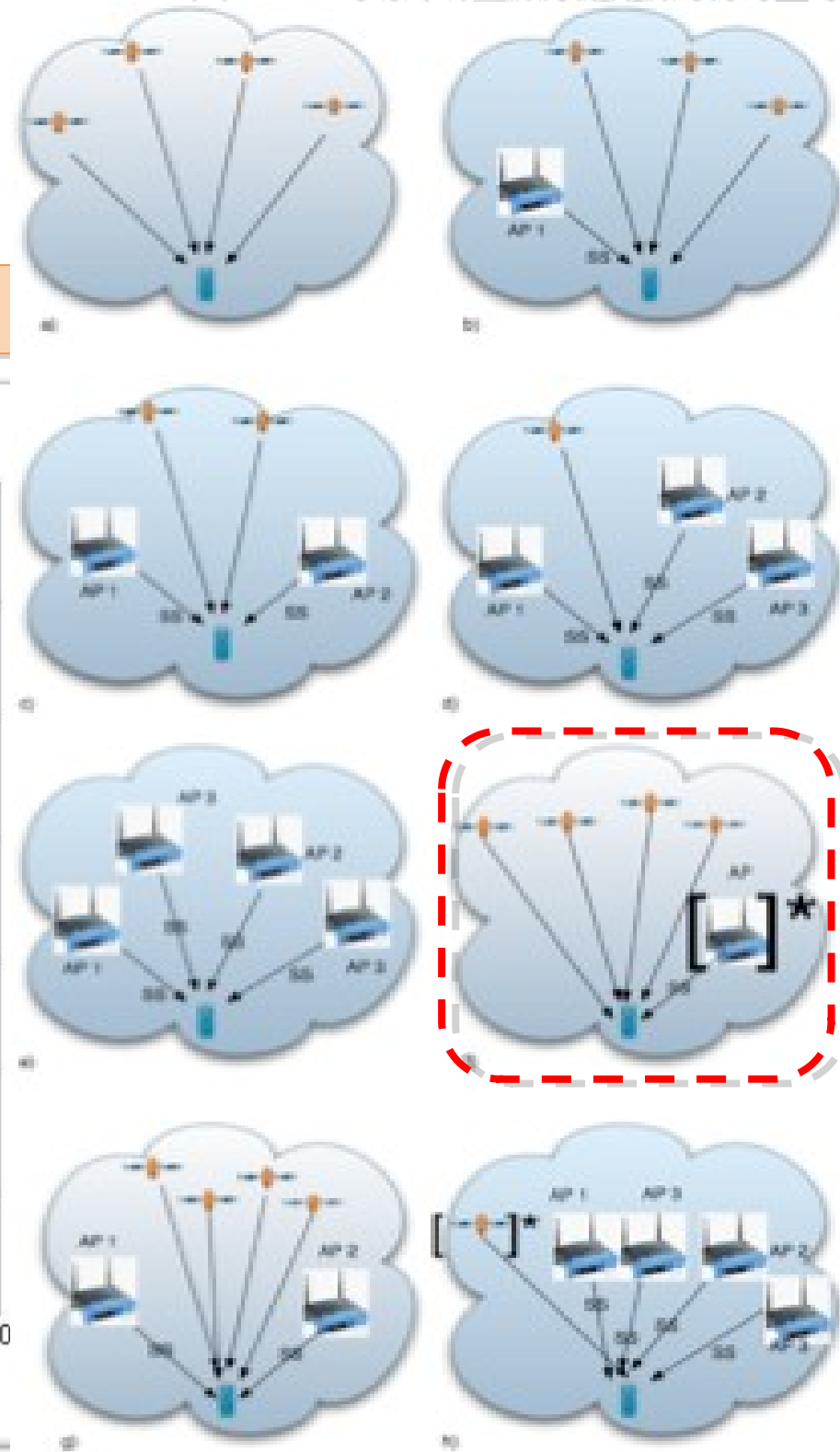
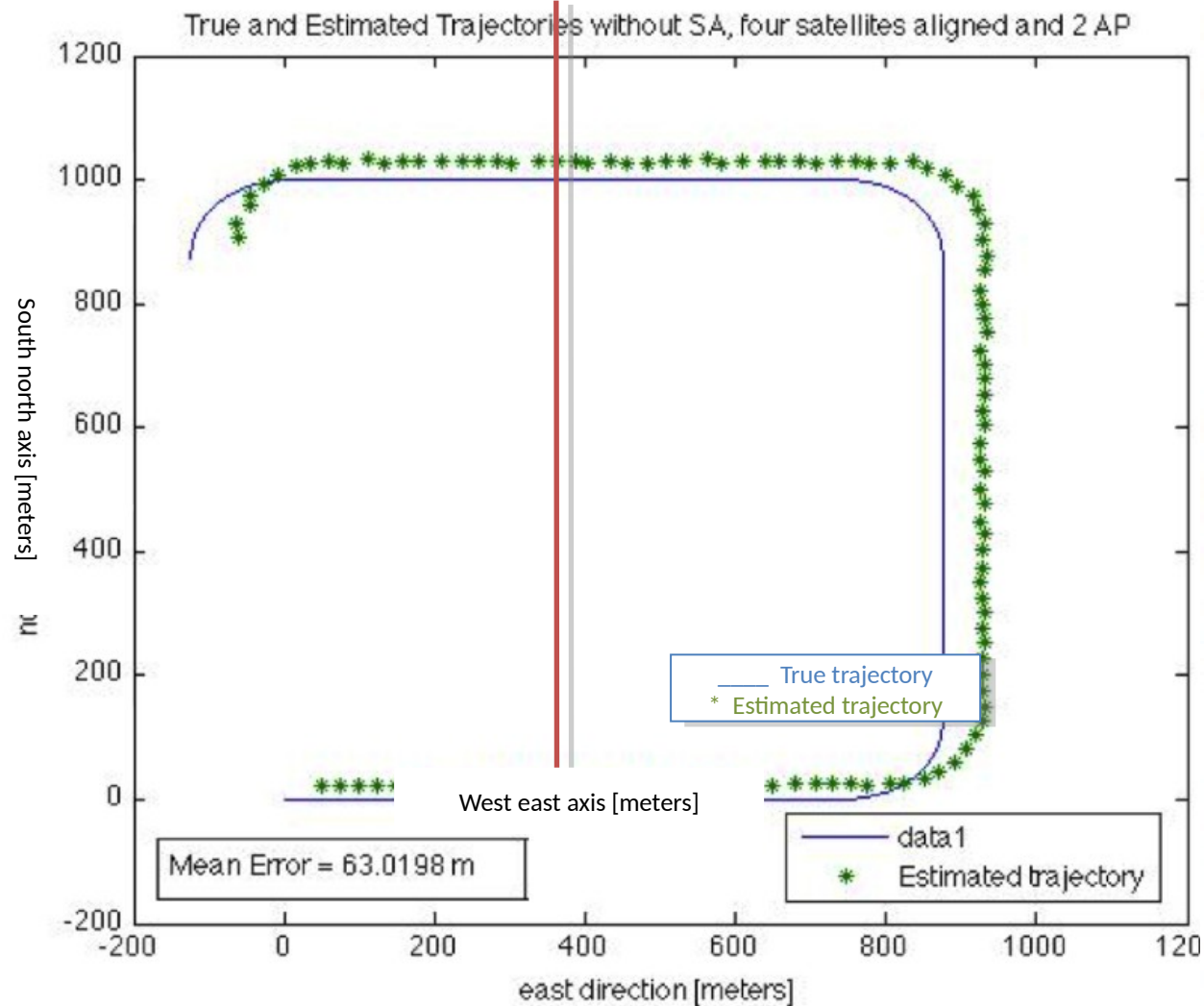
Simulation Results

4 aligned satellites



Simulation Results

4 aligned satellites + 2 Wi-Fi APs

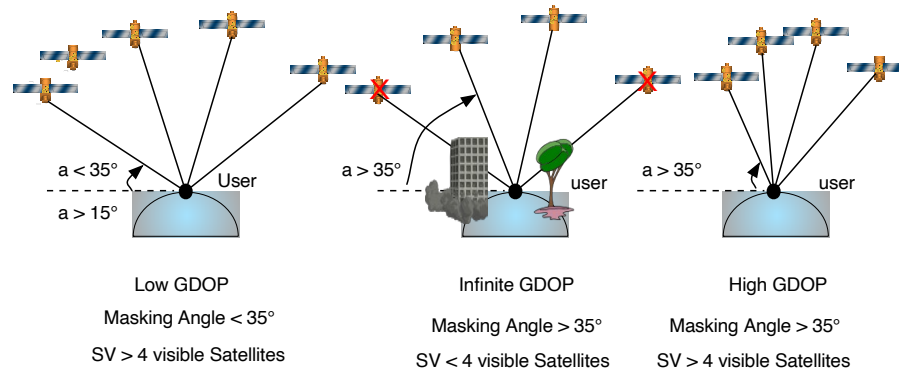


Evaluation of the Repository Qualities

- How to guaranty to improve the accuracy by replacing a satellite by an AP?
- There is a evaluation function of the calculated accuracy without any feedback, this is the *Geometric Dillution of Precision of the GNSS*
- Principle, evaluation of the geometric size of the shape composed of the available satellites and the mobile device
- If the volume is big, the probability to have a good accuracy is high
 - Does not evaluate all the error types, in particular errors which come from the propagation time

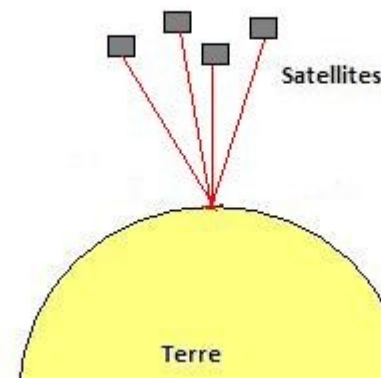
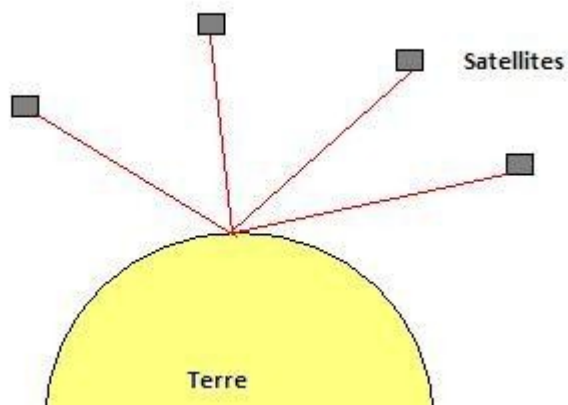
Evaluation : *Geometric Dillution of Precision*

Principle of the GDOP



Good GDOP (1-2)

Bad GDOP (>3)



Conclusions et perspectives



- Improvements still needed on positioning technology such as signal strength and Time of Flight
- Integrate these technologies in combined systems
- Improvements of combination techniques
- Study more combination techniques such as
 - Wi-Fi and Inertial unit
 - GNSS and Inertial unit
 - Wi-Fi and SLAM
 - GNSS, Wi-Fi, inertial unit and SLAM

Variation on Temperature



ReceivedSignal Strength (dBm)	Scenario 13: indoor temperature (number of received packets)	
	<i>23.1°C</i>	<i>18.1°C</i>
-40	0	33
-41	110	358
-42	1335	1619
-43	1377	1545
-44	1253	1094
-45	360	144
-46	38	1
-47	14	0
-48	10	0
-49	7	0
Total	4504	4794
Chi-square	362.1	